

Abstract

The SenseAI project is an AI-driven auditing system developed to transform post-entry inspection processes for customs authorities' transactions in the Kingdom of Bahrain. The system replaces the old papers and Excel auditing that failed to resolve the issues from expanding data sizes, insufficient real-time analysis, and minimal automation capabilities.

A completely serverless cloud-native solution was developed and implemented on Amazon Web Services to overcome these limitations. The solution uses DynamoDB for our scalable data store, AWS Lambda for backend operations, API Gateway for API integrations, and Amazon SageMaker for running all our mission-critical Machine Learning workloads. The system includes a chatbot customized for the use of customs employees, being able to answer customs questions accurately to facilitate the auditing process.

The system features an interactive user interface that enables users to efficiently search and filter transactions, flag and audit them. It includes intelligent risk scoring, employee activity logging, and AI-produced explanations for risk factors that affect transaction scores. SenseAI handled five years of historical customs data with over 13 million transactions. The system enables the exportation of audit results in Excel and Arabic PDF for both technical and management use.

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Chapter 1: Introduction

Auditing of customs is a cornerstone of national revenue protection and compliance monitoring, and effective audit mechanisms are important to the accuracy and efficacy of customs activities. Traditional customs audit techniques, however, are facing a range of challenges in adapting to altered volumes of trade and to new technology. This project sets out a new and innovative method for revolutionizing the customs audit process using the latest AI and cloud technologies to increase efficiency, consistency, and scalability, and aligns with Bahrain's strategic objectives.

This project is the result of collaborative work between the Bahrain Customs Authority, Amazon Web Services (AWS), Tamkeen, and the University of Bahrain Cloud Innovation Center. Creation of an operational prototype that represents a proof-of-concept for the to-be-deployed system is the key mission of this project. Embracing an experimental strategy is one of the CIC's methodologies. One needs to achieve high goals and be bold to ensure the spread of innovation and expansion.

The problem in question is the inefficiency and lack of reliability of manual customs audits. Manual audits are time-consuming and prone to errors, leading to delays and missed revenue opportunities. The lack of a single system to trigger suspicious activity also leads to these issues, making it difficult to have an all-around view of customs operations. The thesis of this project is that a customs audit solution based on AI can dramatically enhance the efficiency, precision, and effectiveness of customs audits in Bahrain.

1.1 Problem Statement

Current customs auditing processes in Bahrain are largely manual, resulting in a number of severe issues affecting the efficiency and effectiveness of customs operations. Manual audits are time-consuming and subject to human mistakes, which not only delays the audit process but also increases the likelihood of overlooking significant discrepancies. This inefficiency is a major concern as it prevents the ability to conduct thorough and timely audits, thus affecting the overall effectiveness of customs operations. According to a Bahrain Customs Affairs report, there is a huge amount of manual work involved in post-clearance audits (PCA), and it leads to delays and inaccuracies (Ebrahim, 2025).

Revenue leakage is another massive problem. The lack of intelligent tracking and analysis of past declarations results in revenues being left untapped. Without an automated process that detects anomalies, revenue from customs duties and taxes remains unrealized. A report indicated that Bahrain Customs discovered BD 250,000 of missed revenues from seaport operations alone in one year. This indicates how wide the gap is in today's audit process, which should be plugged to protect national revenues.

In addition, there is no integrated system at the level of identifying suspicious activity on different points of entry. This fragmentation does not provide for a comprehensive view of customs operations and for identifying patterns of non-compliance. The absence of an integrated system hinders the ability to conduct effective and timely audits of all elements, ultimately resulting in the overall ineffectiveness of customs operations (Customs, 2025).

These are issues that necessitate the development of an automated system to enhance customs audit efficiency and precision. Through the mitigation of these issues, the project aims to modernize customs operations, reduce inaccuracies, and increase revenue collection. The roll-out of an AI-powered system is expected to transform current audit procedures by making them efficient and reliable.

1.2 Project Objectives

The primary objective of the project is to modernize customs audits & enhance the quality, reliability, and security of the audit process in Bahrain. The project aims to align with national standards to foster consistency across customs operations, streamline and automate audit processes to support auditors by providing a user-friendly solution, ensure scalability by enabling seamless adoption across customs entry points, and develop a sustainable solution that minimizes resource consumption.

1. **Automate Customs Audit Analysis:** The project seeks to leverage AI to automate the analysis of customs audits. This involves developing algorithms that can quickly and accurately process large volumes of customs data, identifying discrepancies that would be difficult to detect manually. By automating these processes, the project aims to reduce the time and effort required for audits, thereby increasing efficiency and accuracy.
2. **Provide Dashboards, Alerts, and Pattern Recognition:** The project aims to enhance the ability to monitor and analyze customs data through the use of dashboards, alerts, and pattern recognition. Dashboards will offer real-time insights into customs operations, while alerts will notify auditors of potential issues. Pattern recognition will help identify trends and anomalies in customs declarations.
3. **Help Auditors Focus on High-Risk Cases:** By automating routine tasks, the project aims to allow auditors to concentrate on the most significant discrepancies. This will improve the overall effectiveness of customs audits, as auditors can dedicate more time to investigating high-risk cases. The project will provide tools and insights that help auditors prioritize their efforts, ensuring the most critical issues are addressed promptly.
4. **Local Impact: Revenue Protection for Bahrain:** The project aims to protect national revenue by improving the efficiency of customs audits in Bahrain. By identifying and rectifying discrepancies in customs declarations, the project seeks to recover substantial revenue that is currently lost due to inefficiencies in the audit process. This increase in revenue is expected to have a positive impact on the national economy, supporting various development initiatives.
5. **Ensure Scalability and Reliability:** The project aims to deliver a comprehensive AI-driven solution that addresses these objectives while ensuring scalability and reliability. The solution will be designed to integrate seamlessly with existing customs systems, providing a user-friendly interface for auditors and other stakeholders. By leveraging cloud technologies, the project will ensure that the system can scale to meet the needs of customs authorities of all sizes.
6. **Develop a Sustainable Solution:** The project aims to develop a sustainable solution that minimizes resource consumption. By automating the audit process and leveraging cloud technologies, the project seeks to reduce the environmental impact of customs operations.

1.3 Significance of the project

The project introduces a cutting-edge, AI-driven solution to solve the inefficiencies of legacy customs audits. By automating audit analysis, providing real-time dashboards, and utilizing pattern recognition, the project transforms the current labor-intensive, manual processes into a leaner, more efficient process. This transformation fits into Bahrain's vision of technology-facilitated governance and its national strategic goals. Below is an overview of the significance of this solution:

1. **Efficiency and Accuracy:** The outdated customs audit practices are time consuming, error prone, and inconsistent, leading to most of the times delay in identifying discrepancies. By automating the audit process, the project drastically reduces errors and improves efficiency. AI machines are significantly faster at processing vast amounts of data than human beings, thereby reducing costs and producing more accurate results. This improvement increases the overall quality and dependability of customs activities.
2. **Protection of Revenue:** It is not possible to detect revenue leakage and fraud with ease through the manual process of auditing past declarations. By using automated analysis of historical data, the project enhances the ability to identify anomalies and suspicious transactions. Through this timely detection, national revenue is shielded from being lost by identifying and remedying discrepancies in customs declarations, with more revenue being collected from customs taxes and duties.
3. **Scalability and Innovation:** With the application of cloud-based architecture, the project aims to solve the requirements of large-scale customs processes. The resources are dynamically allocated depending on needs, being cost-efficient and scalable. Any size customs authority, from local offices to national implementations, can leverage the platform. Its cutting-edge application of cloud computing and AI is a demonstration of the extent to which cutting-edge techniques can solve tangible, real-world problems, setting an example for future customs audit solutions.
4. **Support and Collaboration:** The project is a collaborative initiative of government, academia, and leading technology providers, thus ensuring the solution is robust and well supported. Through collaboration, not only is the quality of the solution enhanced, but also its sustainability and scalability are ensured. Through the synergy of the expertise of the involved parties, the project ensures access to the latest research and technological advancements.
5. **Alignment with National Goals:** By automating customs audits as part of its process, the project supports Bahrain's vision of having an effective and transparent government. With the platform in place, every customs declaration is subjected to assessment on equal and unbiased terms, which facilitates offering the customs authority the capability to offer effective revenue protection as

well as monitoring of compliance. This is aligned with the government of Bahrain's strategic objectives of improving the quality and fairness of customs operations throughout the country.

6. **Convergence of Next-Generation Technologies:** The project demonstrates how next-generation technologies like cloud computing and AI can be integrated into government operations. In enabling the preparation of diverse, pertinent, and impartial audit reports, these technologies make customs operations innovative and efficient. The project proves how modern solutions can solve age-old problems in customs auditing through the application of technologies such as Amazon Bedrock, OpenSearch Serverless, and S3. Integration raises confidence in applying AI in broader government uses as well as improving the audit process.

1.4 Report Outline

The report is divided into several chapters, each focusing on a specific aspect of the project:

Chapter 2: This chapter provides an overview of current customs audit practices, highlighting their limitations and the potential benefits of AI-driven solutions. It also reviews relevant literature on AI applications in auditing, providing a theoretical foundation for the project.

Chapter 3: This chapter outlines the methodology used to develop the AI-driven customs audit solution, including the project management framework, development phases, and key milestones. It also discusses the challenges encountered during the development process.

Chapter 4: This chapter describes the techniques used to gather requirements from stakeholders, including interviews, surveys, and workshops. It also presents the system models developed based on these requirements, including use case diagrams and diagrams.

Chapter 5: This chapter provides a detailed description of the system's design, including the user interface, system architecture, and the algorithms used for data analysis and pattern recognition. It also discusses the integration of the system with existing customs systems and the use of cloud technologies to ensure scalability and reliability.

Chapter 6: This chapter describes the implementation process, including the development of the system's components, the integration of the system with existing customs systems, and the deployment of the system on the cloud. It also discusses the testing process, including unit testing, integration testing, and user acceptance testing.

Chapter 7: This chapter summarizes the project's achievements, highlighting the key benefits of the AI-driven customs audit solution. It also discusses the limitations of the current system and proposes areas for future research and development.

Chapter 2: Literature Review

2.1 Introduction

Not only in the Kingdom of Bahrain but the whole world, the rapid growth and advancement of international trade, and the growing complexity of customs regulations, have led to a need for a more easy, efficient, and transparent post-entry auditing process. The traditional approach to auditing is often manual and time-consuming, and unsuitable for identifying complex risks or ensuring full compliance with global and local regulations. This has caused many organizations & government agencies to evaluate if AI, data analytics, and other digital platforms can predetermine or help predict the movement of illegal/contraband goods; however, post-entry auditing is not clearly defined in this context.

This chapter outlines existing literature and technology, as well as relevant industry practices regarding the development of an AI-driven customs auditing system. The literature review describes multiple relevant areas, including innovations in automated auditing, post-clearance considerations to cross-border data flows, as well as examples of applying AI and similar technologies in situations driven by compliance, such as healthcare, and examples of current tools and systems that may inform the development of the proposed systems. This review will allow the project to be situated in the broader context of technological innovation and regulatory considerations.

2.2 Related Research & Practices in Automated Auditing

Automated auditing is becoming an area of interest across many sectors for the improvement in efficiency and accuracy in the auditing process and risk management. Many research and industry-focused projects have taken on the challenge of understanding how AI, machine learning, and data analytics can be applied to support complex auditing tasks generally performed by people. In regard to customs, some international frameworks and case studies discuss the impact and importance of using data-driven post-clearance auditing and automated risk assessment and identification to support customs functions, which helps in fraud detection and accurate decisions. This section will review generally the key trends, chapters, and findings from the literature related to automated auditing practices.

AI has rapidly emerged as a powerful enabler for next-generation auditing. Research has shown that automated auditing frameworks improve consistency, reduce errors, and can proactively detect anomalous financial or trade behavior through predictive modeling (Altundağ, 2024; Talla et al., 2022). In particular, tools like MindBridge and techniques such as Random Forests, Isolation Forests, and

SHAP values have been applied to simulate human-like judgment and uncover patterns in large data volumes.

Current studies have validated the worth of unsupervised models in detecting anomalies in financial and trading audits. Random Forests and real-time inference pipelines were employed in one study to perform automatic compliance detection with high accuracy. A paper discusses the employment of SHAP values in explaining audit model predictions for AI-supported decision-making explainability (Altundağ, 2024). Besides, natural language processing (NLP) integration has also been researched to enable audit documentation automation and the generation of audit summaries from unstructured evidence (Talla et al., 2022).

2.3 Cross-Border Data Flows

Transfer of information between borders is simultaneously a pillar of good trade, a show of conformity, and a regulation of cooperation, as another internationalized trade agreement. Nevertheless, cross-border information flow (CBDF) is now being hurt by technical, legal, and policy issues, especially in the case of the customs authorities, which are relying on cloud computing, machine learning, and real-time anomaly detection. The Internet, as the foundation of international business, allows businesses to control the supply chains, customs declarations, and work outside the accounts that are usually audited and controlled (Meltzer, 2014). The existing liberal movement of information allows trading in real time but demands the creation of international treaties.

International treaties should be implemented to further establish confidence in the CBDF outside the customs office and trade exchange to collectively enforce, assess liability, and keeping of audited accounts. It has, however, been challenging to transfer sensitive data on trade, despite the fact that data localization laws have already been approved in many countries. These necessities characterize the local data storage, processing, and preservation, as opposed to cloud-based audit instruments, such as SenseAI. This is because these regulatory mandates stifle innovation and pose compliance challenges, whereby refined methods may require machine smarts and cloud-based configurations to adequately customize audits (World Monetary Forum, 2020). This type of restriction has a detrimental effect on customs AI systems since it denies access to historical transaction data used in anomaly detection algorithms and hurts Retrieval-Augmented Generation (RAG) systems, which require access to open and interoperable knowledge bases.

The means of getting out of these difficulties are presented by policy researchers in the form of interoperable confidentiality standards and cloud observability patterns. Establishing Mutual Recognition

Arrangements (MRAs) between customs administrations that fulfill the conditions of secured, standardized exchange of audited account information across borders is one of the important solutions. The SenseAI use case architecture stores sensitive data within small areas of AWS with the system's design, and then safely performs metadata operations to calculate anomaly scores. AWS capabilities, such as CloudTrail and IAM, provide the possibility to exercise control over access on a per-object basis and have the full audit capability to allow the customs department in Bahrain to use the system without breaching the sovereignty of data.

Finally, there is an important interdependence of opportunities and threats of contemporary, efficient auditing across borders. Next-gen customs platforms that aim to offer secure, transparent, and digitally advanced trade environments need to meet and tackle compliance limits.

2.4 Healthcare Use Case: RAG, Bedrock, and SageMaker

While customs auditing and healthcare would appear to be operating in extremely different regulatory environments, both have one issue in common: needing to make sense of a lot of structured and unstructured data to determine anomalies, stay compliant, and provide auditable reasons why. The healthcare sector has been a proving ground for the type of AI methods, namely Retrieval-Augmented Generation (RAG) and anomaly detection using SageMaker, that are currently employed in post-entry customs audits.

In one healthcare-focused research study, AWS SageMaker was used to create models that identified anomalies in insurance claims (Murugesan et al., 2024). Those models did not rely on existing fraud rules; rather, they developed patterns of normal behavior through learning past records. If a new claim did not conform to those patterns due to a treatment code discrepancy, unusual cost levels, or provider patterns, it was flagged for further scrutiny. The strength here was that it was flexible: it did not have to be expert-manufactured rules and could learn with the data. An example is the SenseAI system's use of an Isolation Forest to detect anomalies in customs data, where anomalous transactions can be inconsistent HS codes, weight/value discrepancy, or abnormal supplier behavior.

To complement quantitative analysis of anomalies, healthcare researchers also utilized Retrieval-Augmented Generation (RAG) to facilitate explainability and auditability. In a case deployment, a chatbot was developed using RAG to assist health insurance auditors in explaining the policy meaning of emphasized claims (Siddiqui et al., 2024). When an auditor asked why a claim was highlighted as a question, the system brought in applicable parts of health billing codes and rules of compliance and made use of a domain-tuned large language model (LLM) to generate an answer in plain language.

This system not only served as a query engine but also as a context assistant, able to pull in information from heterogeneous documents and adapt the response according to the user's intention. This resulted in SenseAI integration with Amazon Bedrock, with the Nova Pro model being integrated with customs policy documents and regulation handbooks to allow auditors to understand why imports are being raised. In both cases, RAG is the machine prediction-to-human judgment bridge.

Another valuable input is provided by Zhou et al. (2024), who demonstrated that aggregated knowledge models significantly improve the quality of domain-specific Q&A systems. If stacked upon RAG pipelines, the systems performed better and improved the relevance of the responses, even in those domains where terminology and classification were not consistent, a common issue for both healthcare claims and customs HS code classifications. This paper emphasizes the importance of structured retrieval systems and strongly filtered knowledge repositories when leveraging LLMs in audit-related use cases. By looking at how RAG and SageMaker have already been used in a high-stakes, compliance-dense environment such as healthcare, it can be easily seen that the same methods are not only possible but extremely useful in customs auditing. To use SageMaker and Bedrock as the platforms on which to develop SenseAI was therefore based on this analogy: both environments stand to gain from models that can identify subtle anomalies in structured data and describe them with clear, document-backed dialogue. In short, healthcare applications offer a concrete, tested model for what the SenseAI system is trying to achieve: combining statistical detection with human-readable justification, enabling auditors to work faster, bolder, and more transparently than traditional procedures allow.

2.5 Similar Tools and Systems

Numerous tools, platforms, and systems have been created across different industries to support auditing, compliance monitoring, and risk management. These tools, platforms, and systems have provided valuable context on how automation, artificial intelligence, and advanced data analytics can improve auditing processes and decision-making. Examining both unique trade compliance systems and auditing tools in sectors such as finance, supply chain management, and health care will serve our purpose in customs, viewing both bespoke systems and relevant systems of audit. This section features examples of these tools and systems, and an overview of their main functions, strengths, and weaknesses. The information shown in the analysis will inform the design of the proposed AI-powered post-entry customs auditing system, showing best practices and gaps for innovation.

2.5.1 SAP Global Trade Services (SAP GTS)

SAP Global Trade Services (SAP GTS) is a solution that automates and streamlines the trade compliance process to help reduce compliance costs, decrease the risk of non-compliance penalties, and improve customs clearance (SAP, n.d.). The SAP GTS integrates global trade management in one central repository with import and export management capabilities, optimized tariff classification, free trade agreement management, customs management, self-filing, and broker interaction.

Some key functions provided by the SAP GTS include sanctioned party list (SPL) screening, which screens transactions across departmental functions, and blocks transactions pending review that do not pass compliance checks. The SAP GTS provides a more efficient customs management versus other platforms by securing accurate product categorization, determining tariff code classification for invoices when items are imported and exported. Integration with core logistics operations and multiple governmental customs systems supports compliant and efficient management of imports and exports across any flow of goods that requires trade compliance, with the ability to incorporate real-time compliance verification and compliance controls in the order and shipment process. Furthermore, SAP GTS provides support for customs special procedures such as foreign trade zones, bonded warehouses, processing trade in China, and EMCS (the Excise Movement and Control System). Using the SAP HANA database, the SAP GTS application affirms its application data analysis with advanced in-memory computing technology and enhances data modeling, lifecycle management, and security (SAP, n.d.).

AI-driven technology has recently made significant strides in strengthening SAP Global Trade Services' (GTS) ability to manage different elements of complex trade compliance management. The integration of advanced AI technology in SAP GTS is already supporting more efficient, adaptive, and proactive trade compliance management during an ever-increasing volume of global trade, with an ever-increasing complexity (Kommineni, 2020; Talla et al., 2022). To support SAP GTS, functions associated with trade compliance processes have traditionally leveraged static rule-based compliance systems. Therefore, static rule-based compliance systems have been challenged to manage trade compliance processes because of the pace of change of trade regulations and global market conditions. Moving beyond static rule-based compliance systems to leverage AI techniques such as machine learning (ML), natural language processing (NLP), and predictive analytics, SAP GTS extends its capabilities to detect anomalies in real-time, implement suitable regulatory changes in a dynamic fashion, and enable intelligent risk mitigation (Manikyala et al., 2024). A significant development is reciprocal symmetry models, which allow SAP GTS to create dynamic feedback loops that facilitate continuous compliance checks and subsequent changes to monitoring and compliance holistically based upon changed trade behaviors and regulatory changes (Kommineni, 2020; Talla et al., 2022). Systems using this approach are more responsive to changed trade agreements, sanctions, and export restrictions, and they allow organizations to maintain

compliance while enhancing operational agility. Simultaneously, ML-based predictive analytics facilitate predicting potential compliance challenges and deviations, allowing the organization to take more preventative actions instead of exclusively acting reactively (Manikyala et al., 2024).

The employment of NLP in SAP GTS enhances compliance in the sense that it automates a regulatory definition and legal document interpretation process that is fundamentally complex. The complexities of trade regulations and tariff codes offer a more streamlined approach to processing trade regulations and making compliance decisions based upon the most current evidence and regulatory interpretation (Talla et al., 2022). Furthermore, AI-enabled anomaly detection also supports the discovery of dubious transactions and inconsistencies in trade data, which enables improved fraud detection and risk management (Kommineni, 2020; Manikyala et al., 2024).

Another notable feature of the SAP GTS systems involves symmetric trade analytics in discovering trade across multiple dimensional spaces, such as product or service categories, geographic regions, and transaction types. This lens enables organizations to discover discrepancies in trade flows, monitor reciprocal trade agreements, and ensure the compliance and fairness of trading patterns (Talla et al., 2022). In summary, the AI frameworks employed in SAP GTS Civilian extensions deliver additional decision support and autonomy using enhanced dashboards and real-time, contextualized data.

Compliance teams are being empowered in their global risk management capabilities (Manikyala et al., 2024).

Despite these improvements, the research data clearly identifies several difficulties in adopting AI-enabled functionality for SAP GTS, such as the availability of high-quality data sourcing and explainability models, and governing data protocols (Kommineni, 2020; Manikyala et al., 2024; Talla et al., 2022). Ensuring transparency, accuracy, and accountability, in compliance enabled through AI, will limit the ability for organizations to deliver on both expectations and realities as regulations continue to change.

In conclusion, AI-enabled SAP GTS fundamentally changes how organizations will approach compliance in global trade management. The changing operating structures toward an AI-enabled SAP GTS in leveraging AI-enabled operational and predictive analysis, and reciprocal symmetry analysis facilitate the movement away from reactive compliance management to adaptive compliance management that results in enhanced efficiency, agility, and proactive, strategic advantage in local, regional, and global trade management (Kommineni, 2020; Talla et al., 2022; Manikyala et al., 2024).

2.5.2 MindBridge AI Auditor

MindBridge AI Auditor is a leading AI-powered platform to improve how audits of financial records are performed by offering the highest level of quality and efficiency. MindBridge AI Auditor supports the auditor to embrace big data analysis, anomaly detection, risk assessment, fraud detection, and natural language processing, to offer a more comprehensive and insightful audit result (Altundağ, 2024). The platform adds value by allowing the user to analyze volumes of financial records, identify hidden patterns, and potential risks and exposures/irregularities that can so easily be overlooked in traditional auditing practice. In other words, MindBridge AI Auditor takes the Auditors beyond the typical, top-level analysis trend and allows the auditor to identify deeper insights of client risk through full population testing and cutting-edge anomaly detection (MindBridge, n.d.).

A key advantage to the MindBridge AI Auditor is its ability to optimize team expertise through integration and providing, and supportive, professional judgement. MindBridge AI Auditor equips audit professionals to effectively manage complex data and decisions at greater pace and confidence; the platform integrates and minimizes the variation during a contemporaneously audit, via promoting consistent and objective analytical paper trails (MindBridge, n.d.). Every engagement is based on the same data-driven basis, across the entire audit engagement. MindBridge AI Auditor promotes efficiency and marks the start of optimization for offering the services with less call on resources in some cases (e.g., expense validation), and milestone insights in the case of more complex and higher risk analyses. MindBridge AI Auditor offers its user a sophisticated data driven risk assessment component that can be used to inform financial risks, and offer better audits through more effective resource allocations - to have meaningful allocated audit staff cases. Also offers fraud detection to help organizations proactively deter possible losses through suspicious transactions (or similar). This analytic capability is scaleable, and can be used without sophisticated tools or without a big-data analyst (Altundağ, 2024; MindBridge, n.d.).

Going beyond its technical development, MindBridge AI Auditor provides demonstrable business value. It make the organizations capable of enhancing their audit quality, eliminate work repetitions, and significantly reduce the time to complete engagement work cycles. Adopting MindBridge allows audit firms to differentiate themselves through innovation and showing their commitment to transparency and enhancing and increasing the value of their brand. In addition to that, it can also improve the attraction and retention of talent by adopting modern tools that reduce the number of repetitive tasks, allowing professionals to focus on higher-value work (MindBridge, n.d.).

The successful deployment of MindBridge AI Auditor across various international markets (especially

throughout the United States, Canada, and United Kingdom) indicates the potential relevance and even effectiveness of its application in the modern audit environment (Altundağ, 2024). The capabilities and lesson learned from the implementation of MindBridge AI Auditor - i.e, explainable risk scoring, automated anomaly detection, and coupling of AI with human auditor workflows - can provide helpful guidance for the establishment of AI-endorsed post-entry customs auditing systems.

2.5.2.1 MindBridge AI's User Stories

Beyond the core technological capabilities, MindBridge AI Auditor has also illustrated its value in the real world with various client success stories throughout the auditing profession. They have scaled these advanced analytics and enhanced audit quality and have documented many (successful) case studies for publication on MindBridge's website. These showcase how audit firms and finance teams are embracing and applying advanced technology and analytics to drive transparency, efficiency, and maximum value for their clientele (MindBridge, n.d.).

KPMG, for example, has partnered with MindBridge to showcase how leading audit firms are investing and using AI-driven technology to transform audit practice. KPMG will be able to provide consistent and high-quality audit experience across the globe by embedding MindBridge's next-generation transaction and risk identification capabilities into its audit process (MindBridge, 2024). MindBridge allows KPMG auditors to analyze entire datasets, zero in on the institutions' high-risk areas, and spend time on discretionary activities. MindBridge is aligned with KPMG's operational audit approach as MindBridge incorporates a risk-based, exception-based, and outlier-focused audit approach to highlight and remove routine transactions, where possible, to highlight the more meaningful anomalies and high-risk exceptions for review.

Through its relationship with MindBridge, KPMG will incorporate MindBridge's statistical and machine learning and rules-based analytics into KPMG Clara's own proprietary smart audit platform. This will allow auditors to automate the analysis of larger transaction populations while increasing the firm's overall transparency, explainability, and audit quality. KPMG's use of MindBridge improves audit efficiency and audit quality through enhanced engagement with the client; the firm uses MindBridge to extract more evidence from the population, while the client receives an audit of greater depth and extended coverage. KPMG believes a MindBridge approach to audits is an effective enabler for delivering on its longer-term strategy for delivering the audit of the future to its clients while ensuring its professionals can maintain standards necessary to keep pace with increasingly higher demands from clients and regulators alike (MindBridge, 2024).

KPMG is not the only chartered firm using the MindBridge platform to drive innovation in audit and assurance (MindBridge, n.d.). Other illustrative examples include:

- Baker Tilly Canada used MindBridge to enable its audit teams to move to a data-first audit approach, which allows auditors to reduce the required on-time manual effort needed for risk assessment, while providing deeper insights to the audit.
- Cherry Bekaert LLP used MindBridge to integrate AI-powered analytics into 100% of client data to drive up quality and efficiency and instill greater confidence on behalf of clients.
- CLA (CliftonLarsonAllen LLP) capitalized on the MindBridge platform to provide greater depth and transparency in audit findings and generate the ability for auditors to provide greater value to its clients and stakeholders.

These user stories reflect how artificial intelligence-driven audit platforms like MindBridge are shifting the profession to provide auditors with better tools and achieve better client outcomes across industries and firm sizes. (MindBridge, n.d.).

Chapter 3: Project Management

3.1 Process Model

In developing the SenseAI project, we used a combination of approaches that were complementary and suggested by experts in the AWS team; we initially worked within a backward design approach during the planning and design phase and then an Agile software development methodology during the implementation phase. These approaches provided us with an easy way to maintain our focus on the project goal while ensuring flexibility and adaptability in the development of all the features.

3.1.1 Backward Design methodology (Planning and Design Phase)

At the very beginning of the project, we utilized a backward design approach to help us decide regarding system planning and architecture based on the desired outcomes of the Kingdom of Bahrain customs team. In order to fulfill this goal, we created and defined the key outcomes we needed to support. With the Customs Authority team, we developed user stories that described the expected behaviours of the system, user interactions and responses, as well as the business processes we wanted to ensure the system supported. These user stories provided much-needed clarity in developing what was expected by the end users and ultimately by the client.

From the desired outcomes and user stories that we had prepared, the team then worked backward to understand and conceptualize the key components of the system, the architecture, and the features that would enable the system to deliver what is intended. This method made sure that the whole project was on track at all times to deliver real value based on the needs of the client, not based solely on the technical aspects.

3.1.2 Agile Methodology (Development Phase)

After the planning phase was completed, an Agile software development methodology was implemented for the implementation of the system. In cooperation with AWS supervisors, the team decided to follow the Agile methodology because it offered a workflow that was suitable for a project that requires client engagement and many chances to iterate. An agile development process that consists of sprints with the goal of an agreed-upon set of system features, divided into each sprint based on the priority of the feature.

During each sprint, we facilitated planning sessions, where we attempted to pick and estimate a set of user stories that prioritized use case features in the product backlog to support a focused sprint backlog for the team to work on its tasks. Also, we emphasized collaboration between the development team; we conducted daily Stand-up meetings to monitor where we were in delivering a feature set to ensure

transparency, solve issues, discuss any blocks to the work, and communicate with the Customs team for any clarifications or document requests.

At the end of the sprint phase, review meetings were organized with the Bahrain Customs Authorities team for feedback on tasks done during the sprint. This feedback allowed us to apply adjustments to the product backlog and backlog priorities in GitHub to be picked up in future sprints and to keep pace with changing client requirements.

After the sprint review, an internal retrospective meeting was an important step to allow the development team to reflect on the sprint phase. The comments received would be useful to help strengthen work in the team for future sprint phases and improve workflow with a view to a more successful outcome.

This Agile methodology allowed us to incrementally deliver system features, collaboratively engage with the client, and accommodate new requirements and feedback during the lifecycle of the project.

3.1.3 Process Model Summary

The backward design approach and agile mechanism formed an effective way to manage the project aimed at success. The backward design focused our efforts on building a system architecture that supported the Customs' intended use of the system, and the agile methodology provided the flexibility and iterations to manage the implementation with a committed client interaction presence that allowed us to promote new ideas and adjust changes during the development lifecycle.

3.2 Risk Management

During the project timeline, specifically in the planning process, the team identified many potential risks that could put the successful delivery and core functionalities of the system at risk. The team of identified and done their best to create a mitigation plan for each risk to reduce the likelihood and impact of the risk, and are listed below, the potential risks to our project.

1. Team Member Leaving the Project

- **Description:** The SenseAI team consists of 5 members and a Tech Lead from AWS. The leaving of a team member like a Frontend or Backend developer, PM, or system architect can have a bad impact on the project timeline and productivity of the project. It may also negatively affect team morale, add unnecessary workload to team members needlessly, and ultimately lead to a delay.
- **Level:** High
- **Mitigation:** Encouraging proper documentation in a file named "OUR_readme" within the parent directory of the project, regular internal reviews, weekly working sessions, systematic handover processes, and cross-training to minimize the transition and impact can help mitigate.

2. Unauthorized Access or Data Breach

- **Description:** The customs data carried through SenseAI involves quite a lot of sensitive data. Any unauthorized access or breach could have a significant impact on operational effectiveness, regulatory compliance, and public trust in the system.
- **Level:** High
- **Mitigation:** Following best practices in AWS Identity and Access Management (IAM), activating Multi-Factor Authentication (MFA) once the system is delivered to Bahrain customs authorities using AWS Cognito user credentials, activating detailed logs and activity audit trails, and conducting comprehensive security assessments will allow for work to develop additional controls to eliminate these risks.

Of all these mitigations, encryption is the main key and must be enforced throughout the whole system to protect from data breaches.

3. Insider Threat

- **Description:** There are risks that can also come from internal users that include misuse or damage, which could thus result in data integrity compromise, data loss, and damage to trust, success, or care. As an internal user, your actions may be perceived on some level as malevolent; however, regardless of whether you act with harmful intentions or harmlessness, these internal risks should be regarded as serious ones.
- **Level:** Low
- **Mitigation:** conduct verifications such as for each user in the system, confidentiality agreements, full activity logs, based upon roles and geographical limits on users, and relevant security awareness and compliance training.

4. Limited Availability of AWS Services in the Bahrain Region

- **Description:** There is a restricted number of AWS and Critical Services relevant to the project (Amazon Bedrock, SageMaker), which may create limitations in system capacity and could cause inefficiencies or complications in system production.
- **Level:** Medium
- **Mitigation:** building and deploying SenseAI architecture can be modified to allow cross-regional deployment and the ability to use the available options in other regions, e.g. anchoring to Ireland (eu-west-1), and apply those services consistent with the regulatory requirements established in Bahrain IGA, to therefore allow continued operational integrity and privacy of data.

5. Bugs or Performance Issues in AI Model (SageMaker Risk Scoring)

- **Description:** There is the possibility of some failure or performance, or accuracy issues with AI models, more specifically Amazon SageMaker, which is a critical component of the risk scoring. The above subset of issues can pose a material risk to the system's reliability and its accuracy, and subsequently, user confidence in the accuracy of the system deliverable.

- **Level:** Medium
- **Mitigation:** Implement ordinary basis monitoring and validation procedures for AI models, implement procedures for validating AI model fidelity and performance before and after execution. Keep agility in your process so that there can be fast retraining and fine-tuning based on performance indicators and user feedback, which aids in getting ahead of issue resolution as soon as possible.

6. Errors in Uploaded Data That Compromise System Accuracy

- **Description:** Uploading data that is inaccurate or incomplete or in poor formats could lead to serious errors for the system, which will have serious implications for data integrity, trustworthiness, and the overall reliability of the entire operational workflow.
- **Level:** Medium
- **Mitigation:** Improved verification frameworks and smaller previews and edit functions at the time of upload on the upload screen should reduce the likelihood of experiencing any issues that arise from the data upload phase.

7. Changes in Customs Regulations or Operational Processes

- **Description:** Changes to customs regulations or operational processes can lead to suggested or required changes to the system very quickly or unexpectedly, which could inhibit the implementation of the system. Some changes may require substantial efforts to redevelop and test them.
- **Level:** Medium
- **Mitigation:** Aim to have components of the structure and a dynamic Knowledge Base that could be used to adapt to new operational processes or legislative changes quickly. Trained personnel to continuously identify changes and communicate needed actions in a precise.

8. System Degrade/Crash with High Volume of Data (greater than 13 million transactions).

- **Description:** Given the high volume of transactions, there is a very real risk of crashing the system, slowdowns, or other poor performance that would impact operational performance, operational stability, and reliability. Keeping the level of performance and up-time would be necessary for reliability to conduct auditing processes on these transactions as intended.
- **Level:** Medium
- **Mitigation:** Use a sized and scaled sized and reliable serverless AWS architecture, complete with load and performance testing, structured testing, continual performance monitoring, alerts, and optimizations, effective methods (e.g., batching, paging) into the process's workflow.

9. Model Drift (SageMaker Risk Scoring)

- **Description:** The accuracy of the SageMaker risk scoring model can possibly drift over time because of some combination of changing transactional patterns, data

distribution, or workflow changes that would impact the accuracy and reliability of risk assessment operations.

- **Level:** Low
- **Mitigation:** Establish a process of continuous evaluation of SageMaker scripts and retraining against routinely reviewed datasets, continuous operational improvement feedback, and predictive analytics to identify and eliminate model drift.

10. Lambda Limits (Timeout, Memory, Payload Size)

- **Description:** Limits around AWS Lambda (time of execution, memory used, payload size) can stop the progress of the data flow within a workflow, cause heroic inefficiencies, or degrade the user experience.
- **Level:** Low
- **Mitigation:** Validate upload file sizes, create Lambda functionality that is efficient with memory and processing, monitoring, alerting, and resilient retry situations, will be best at driving workflow advancement.

11. AI Chatbot Hallucinations (Bedrock Knowledge Base)

- **Description:** The AI chatbots may deliver misleading or conjectured information if they have not attended to and/or adequately curated sufficient components of the Knowledge Base that would lead to misinformation or confusion.
- **Level:** Medium
- **Mitigation:** Utilize prompt engineering best practices, conserve the Knowledge Base and continue to add to and regularly update, clearly mark AI content generated by any chatbots for all intents and purposes, and train users on proper use and engagement which can limit omissions and also generally provide a better experience overall.

The following table summarizes the key risks considered, sorted in descending order:

Risk	Level	Mitigation Strategy
Team Member Leaving the Project	High	Documentation, internal reviews, handover processes, and cross-training.
Lambda Limits (Timeout, Memory, Payload Size)	High	Validate file sizes, efficient Lambda functions, monitoring, and resilient retries.
Unauthorized Access or Data Breach	High	IAM best practices, MFA activation, detailed logs, audits, and encryption.
AI Chatbot Hallucinations (Bedrock Knowledge Base)	Medium	Prompt engineering, update Knowledge Base, mark AI content, and user training.
Limited Availability of AWS Services in the Bahrain Region	Medium	Cross-regional deployment, deploying in the Ireland (eu-west-1) region, and ensure Bahrain IGA compliance.

Bugs or Performance Issues in AI Model (SageMaker Risk Scoring)	Medium	Monitoring, validation, retraining, fine-tuning, and user feedback.
Errors in Uploaded Data That Compromise System Accuracy	Medium	Verification frameworks, previews, and editing capabilities.
Changes in Customs Regulations or Operational Processes	Medium	Dynamic Knowledge Base, trained personnel, and adapt to changes.
System Degrade/Crash with High Volume of Data (greater than 13 million transactions)	Medium	Serverless AWS architecture, testing, monitoring, batching, and paging.
Insider Threat	Low	User verifications, confidentiality agreements, logs, and training.
Model Drift (SageMaker Risk Scoring)	Low	Continuous evaluation, retraining, operational feedback, predictive analytics.

Table 1. Table of risks

3.3 Project Activities Plan

To achieve efficient system development and delivery, project activities were scheduled according to a structured timeline. Tasks are assigned to particular time frames, allowing for systematic progress, accurate tracking, and timely completion of each project phase.

Pre-Activities:

- Program introduction.
- Workshop Session Covering AWS Services.
- Working Backwards Philosophy Press Release/Frequently Asked Questions(PR/FAQs).
- Relevant workshops about Git and Cdk.

Weekly Activities:

- Monday and Thursday, 3-hour working sessions to develop and collaborate with each other and our CIC advisor.

Biweekly Activity Agenda (Sprint planning/ Retro and Demo session):

- Demo Showcase: Showcase the progress of the PoC development to the sponsor.
- Retro Session: capturing what went well/ what needs to be improved.
- Sprint Planning: Planning activities for the upcoming sprint (2 weeks).

Activity 1: Initial Requirements Gathering and Analysis

- Date: 17th of march.

- Summary: Conduct a meeting session with the stakeholders, the customs affairs, to gather and analyze requirements. Document the user stories and classification by priority.

Workshop Activity: Introduction to Agile Project Management

- Date: 7th of April.
- Summary: A brief introduction to the flow of agile project management, the team will follow, integrating GitHub projects, a priority backlog tool, and the concept of user stories. Also involved creating tasks based on the user stories, choosing which user stories to cover in the current sprint, and task distribution.

Activity 2: Demo, Retrospective, and Sprint Planning

- Date: 28th of April.
- Summary: Showcase the demo centered around user story 1. Planning for the next sprint, organizing the tasks as issues in the GitHub project backlog.

Activity 3: Demo, Retrospective, and Sprint Planning

- Date: 8th of May.
- Summary: Showcase the demo centered around user stories 2 to 15. Conduct a retro session for a few minutes. Plan for the next sprint by organizing the tasks as issues in the GitHub project backlog.

Chapter 4: Requirement Collection and Analysis

4.1 Requirement Elicitation

To guarantee that the SenseAI solution effectively solved the real-world issues of auditors within the Department of Customs and met the strategic aspirations of the Bahrain Customs Authority, an intensive and iterative requirement elicitation process was carried out. The development team employed various methodologies to elicit requirements, determine functional expectations, and record user-centered needs. This process was carried out in collaboration with stakeholders, employing the AWS Cloud Innovation Center (CIC) methodology, specifically the "Working Backwards" process, which begins with the end-user experience and frames success in terms of user outcomes.

One of the key methods used in this project was an extensively structured solution workshop that convened customs officials, audit managers, AWS cloud architects, and University of Bahrain developers. The workshop acted as a focal point for brainstorming in unity, analysis of problems, and the creation of innovative solutions to the prospective development of customs auditing procedures. By utilizing the Working Backwards method, the workshop participants took part in an exercise where they envisioned the accomplishment of the SenseAI system deployment.

They collaboratively drafted a fictional press release and FAQ to define what the product would achieve if it were already completed and deployed. This backward design approach helped clarify which features were most valuable to users and which weaknesses needed to be resolved. For instance, auditors highlighted the urgent need for automatic risk detection, an interface that could sort and flag transactions based on risk levels, and the ability to see similar past cases to guide decision-making. Leadership participants emphasized the importance of high-level dashboards that would show performance by port and offer numeric and visualized insights into compliance trends over time.

In parallel to the workshop, a series of individual interviews were undertaken with customs officers and senior administrators. Semi-structured interviews provided the opportunity to collect more personal and detailed comments. Although the workshop setting facilitated the ability to recognize collaborative requirements, the interviews brought to light individual frustrations, certain work practices, and particular issues that auditors face on a daily basis. For instance, a number of auditors described how the existing manual system forced them to go through scores of declarations with no automation, frequently having to

just depend on their own experience and instincts to identify fraudulent patterns. Supervisors complained about the lack of consistency among various ports in the selection and execution of audits. Meanwhile, department heads expressed a desire for standardized audit procedures and reliable data to make performance comparisons across regions. These insights informed not only what features the system should have, but also how it should behave, what workflows it should support, and how it could fit seamlessly into the daily routine of its users.

In addition to the structured workshops and interviews, the student development team held several discussions with the AWS supervisor and customs staff to refine the specifications and validate their alignment with real-world customs auditing practices. A milestone accomplishment within this project was achieved during a visit to the Bahrain Customs head office on 20 March 2025, where additional operational input was received. It was stated that the audit unit usually obtains and examines shipment data about a year after clearance, pointing out the necessity of retrospective analytical capacity in the system. Customs administrations suggested during the discussion adding a visual risk indicator on a color-coded scale or percentage-based index to represent the probability of a shipment being non-compliant, taking into account value, HS code, and weight. The significance of examining variances in weight and stipulated item classification of shipments was highlighted as one of the methods of detecting possible abuse of HS codes. Stakeholders also recommended the addition of a function of prioritizing cases involving companies with a past record of offenses, as well as the addition of sophisticated filtering functions to improve audit efficiency. Discussions also included the use of formulas, such as calculating the unit price by dividing the declared value by shipment weight, as a method of detecting irregularities. Furthermore, customs officers expressed interest in automating the comparison between port-level shipment descriptions and official HS code definitions, possibly through external APIs. Lastly, it was proposed that a chatbot component could be enhanced with intelligent recommendations and statistics derived from historical audit data, enabling more contextual and responsive user interaction.

In addition to these exercises, a comprehensive review of current customs documentation was undertaken. The audit team assessed a sequence of customs declaration spreadsheets, hand audit records, and audit report template formats used in various ports in Bahrain. This review facilitated a more significant comprehension of the considerable data points auditors focus on during their decision-making process. It also revealed inconsistencies in data entry methods and formats, emphasizing the need for strict input validation at the point of file uploads. By identifying HS codes that are prone to frequent audits, trends related to repeat offenders, and variations in reported values, the team was able to come up with a data model that could support AI-driven analysis. The review of documentation emphasized the key role of

metadata, i.e., fields such as country of origin, product description, and declared value, in transaction risk determination, which in turn affected the analytical framework of the system.

Parallel to the requirement elicitation activities, a set of user stories was co-created in workshops and subsequent meetings with the stakeholders. The following user stories were developed according to the results of the well-organized workshops, interviews, and the analysis of the documents, and mirror the underlying needs and priorities stated by auditors and leadership teams. These user stories formed the basis upon which the development team designed and developed the SenseAI system to be used in supporting the customs auditing processes in the real-world environment and to ensure that the final product delivers the maximum value to its targeted users.

User stories

User stories are a foundational element in agile software development and system design. They are brief, user-centered descriptions of a feature or requirement, usually framed to highlight the user's role, their desired action, and the intended benefit. This method ensures that each system capability is grounded in practical user needs and contributes directly to the system's usability and value.

In the context of our project, user stories played a crucial role during the initial stages of system design. We conducted interviews and feedback sessions with the Customs Affairs team to collect a broad range of user requirements. These user stories provided insight into the daily tasks, challenges, and expectations of the end users, which helped us design functionalities that are not only relevant but also practical and efficient in their application.

Furthermore, user stories served as a bridge between technical and non-technical team members, facilitating better communication and alignment across the front-end and back-end teams. By documenting user needs in a structured yet accessible way, we were able to prioritize development tasks, structure our sprint planning, and ensure that the final product aligns with real user expectations. The use of user stories contributed significantly to building a more intuitive and user-centered system architecture.

As a [role]	I want to [goal]	So that [benefit]
As an Auditor	I want to conduct audit analysis	So that I can identify discrepancies and improve compliance and identify long-term patterns
As an Auditor	I want to receive pop-up suggestions and recommendations	So that I can take informed actions efficiently.
As an Auditor	I want to generate targeted reports on high-variance transactions	So that I can focus investigations on the most significant revenue discrepancies
As an Auditor	I want to view a dashboard with high-risk transactions alongside a risk percentage	So that I can focus investigations on the high-risk transactions
As an Auditor	I want to view audit logs	So that I can stay updated on important actions and changes
As an Auditor	I want to receive alerts for suspicious declarations.	So that I can immediately flag high-risk shipments for inspection.
As an Auditor	I want to view historical transaction patterns for similar shipments.	So that I can compare current declarations against past patterns to identify anomalies.
As an Auditor	I want to analyze both new shipments and previous shipments	So that I can perform analysis based on the requirements I have received
As an Auditor	I want to generate a report that will show the details of an audited transaction	So that I can see the productivity of employees and the mistakes they made and identify any improvements to the process
As an Auditor	I want to upload a sheet of transactions	So that they can be analyzed and audited by the system
As an Auditor	I want to add business rules	So that I can expand detection criteria
As the Customs Authority Leadership	I want to access comprehensive compliance/revenue analytics across all ports.	So that I can make data-driven decisions for national revenue protection
As the Customs Authority Leadership	I want to assess the distribution of high-risk shipments across different entry points	So that I can optimize staffing and resource allocation across all ports
All Roles	I want a support page/chatbot page (for general inquiries)	So that employees can make general inquiries about Customs

Table 2. User Stories Derived from Requirements Elicitation Activities

Collectively, these elicitation techniques formed a solid foundation for the project. The combination of collaborative workshops, one-on-one interviews, and document analysis allowed the development team to gain a comprehensive view of the users' needs and expectations. This integrated approach guaranteed that the system specifications of SenseAI were not founded on speculation, but rather on a direct, educated

grasp of the customs audit ecosystem. Consequently, the project moved forward with a well-defined and reasoned set of goals, thereby ensuring technical implementation consistency with the end-user impact.

4.2 System Requirements

This section describes the fundamental system requirements of the SenseAI Customs Audit Platform, divided into functional and non-functional. The requirements here were gathered based on stakeholder interviews, customs authority workshops, and observations in the system developed in the project lifecycle.

4.2.1 Functional Requirements

Listed below are the fundamental functionalities that characterize the behavior of the SenseAI system according to its primary purpose: to help streamline post-clearance audits using AI-supported data and insights.

1. **Smart Transaction Upload and Parsing:** The system allows an auditor to upload audit declaration spreadsheets in .xlsx or .csv format. Upon upload, the system automatically identifies fields such as HS code, shipment weight, declared value, and description for each transaction. The data is validated and pre-processed before being sent forward for downstream analysis through machine learning models.
2. **Violation Risk Score and Color Indicator:** The platform reviews each transaction uploaded and supplies a violation risk score on a scale of 0 to 100. This score is complemented by color indicators (e.g., green = low; orange = medium; and red = high) indicating how likely the auditor should prioritize the transaction for audit risk based on violations or misclassification.
3. **Anomaly Detection Based on Historical Patterns:** SenseAI analyzes previous customs data and audit cases to identify irregularities in newly uploaded transactions. For example, if the declared value is low for a shipment of mobile phones or the weight doesn't match what would normally be anticipated, SenseAI takes note of it and pulls any similar historical breaches. This allows for evidence-based audits and expedites the decision-making process.
4. **AI-Powered Chatbot for Policy and Transaction Assistance:** The system offers an integrated AI chatbot solution that supports customs staff by answering questions related to customs policies, regulations, and transaction information. Recognizing the value of an interactive user experience, users can interact with the chatbot to ask about audit follow-ups, what regulations say on particular topics, or ask about HS codes, common violations, or prior transaction patterns. The chatbot utilizes

the historical data contained in the system as well as the documentation built into the system to develop an intelligent response to the user that is relevant to the context. The AI chatbot serves as a repository and makes institutional knowledge more accessible and contextual to enhance timely and relevant decision-making in real-time, especially with regard to audit functions.

5. **Automated Report Generation:** The system provides auditors the ability to produce downloadable reports that summarize key findings of the audit. The reports contain high-risk transactions, risk scores, flags, and notes made by the auditor. For record-keeping purposes, the reports can be exported as a PDF and shared with other staff.

4.2.2 Non-Functional Requirements

Besides the core functionality of the system, it adheres to multiple non-functional requirements in order to meet usability, security, scalability, and performance objectives.

1. **Security:** All data uploaded to the system is AES-256 encrypted both in transit and at rest. Role-based access control is implemented to restrict unauthorized data uploads or submissions/notifications of audit results only to authorized users. CORS headers will restrict any unauthorized third-party API calls from other domains through Lambda functions.
2. **Performance:** This system will also ingest and process large data sets, including 5-year transactions of customs activity. As we will be processing transaction files that often have thousands of rows, the system will either upload and/or process in a very short timeframe. The system has also been designed to operate efficiently during batch audits.
3. **Scalability:** This system uses AWS services such as AWS Lambda functions, AWS S3, and API Gateway. The system has been developed to automatically scale, as required, to the workload demands. Whether one auditor is engaging with the platform or an entire port, there is no restriction on service performance because no manual server configuration is required.
4. **Usability:** The front end has been developed with ReactJS, which provides the system with a clean and responsive UI. Bootstrap has been included in the application's style as conducive to its component consistency and visual aspects, such as scoring, which in some cases has color codes. This makes it user-friendly even for non-technical users who may be looking at audit results visually in summary.
5. **Auditability and Traceability:** Each transaction processed is logged with a timestamp and associated audit decision. This allows for reviewing previous audits, tracking user actions, and

ensuring accountability. The system supports transparency by storing logs in a DynamoDB table for future reference.

4.3 System Models

4.3.1 System architecture

System Flow Diagram:

The architecture and flow of the SenseAI system are described below. It defines the interaction between users with the system parts, as well as the interconnection between various modules and services involved (authentication, data storage, AI processing, and user interfaces) to provide a fluent experience of auditing and reporting.

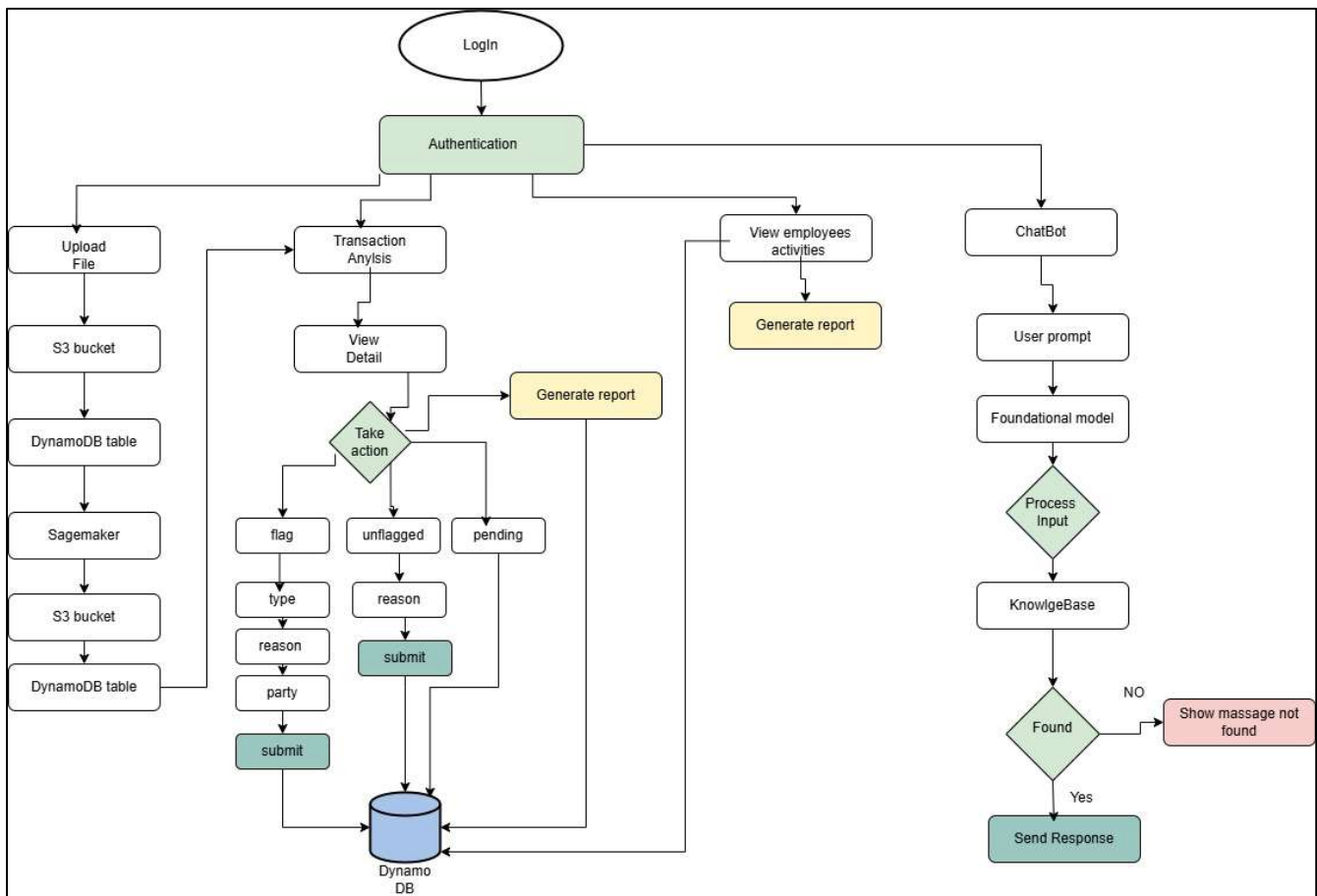


Figure 1. Data Flow and user navigation of SenseAI system

After the user signs in to the system, it starts by performing an authentication process to confirm the identity of the user with the help of AWS Cognito, so that only authorized users can further access the process. A successful login will land the user on different system functionalities, based on the role and permission. The File Uploading module is one of the main components, and the user can only upload

transaction files after choosing a month and year. The system will guarantee the integrity of data because it will not allow duplication of uploads, and there will be audit trails.

The uploaded data is stored in a secure place and made accessible for processing by artificial intelligence (AI) models and visual dashboards. The Auditing Page is the main working space of auditors, where they can see visualizations of the transaction trends, risk scores, and metrics to act on. Here, auditors may drill down to the Transaction Detail View, where they can inspect individual transaction data and examine risk factors, as well as take actions like flagging or annotating transactions that are logged in the activity logs of the system.

The Employee Activities Page helps to support operational transparency by giving administrators a chance to follow and observe employee activity throughout the system. The activity data is kept in a special database table and can be filtered and exported to internal reports and auditing purposes. Also, the system will provide a Chatbot Interaction Page, where the user will interact with an AI-based assistant to get fast responses about customs regulations, procedures, and transaction-related questions. The chatbot is powered by a knowledge base and essential models to deliver contextual answers, which can ease the user support process and workload.

The file storage (S3), data processing (SageMaker), transactional and activity data storage (DynamoDB), user authentication (Cognito), and AI services are all connected in a serverless architecture that is built to provide scalability, flexibility, and traceability of the entire SenseAI system.

4.3.2 Entity Relationship Diagram

The SenseAI system relies on serverless, NoSQL Amazon DynamoDB, which offers elastic and highly scalable data storage. DynamoDB does not impose the strict table relationships or employ an entity-relationship model like in the case of traditional relational databases. It does not; rather, it employs a schema-less design in which individual items (which correspond to rows in a relational database management system) may have varying sets of attributes.

The SenseAI system uses two fundamental entities to assist system capabilities:

The Transaction table contains all the bulk data regarding the customs declaration and audit-related data points, which becomes the basis of risk analysis and AI-driven auditing.

The EmployeeActivities table captures events of employee actions within the system, which can be used to track their activities, auditing of system usage, and provide reports about user behavior.

In order to shed light on the data model, the following tables show the structure and some important attributes of these two major tables.

Table Name: <i>Transaction</i>
Primary and Sort Keys: <i>PK: Reference Number</i> <i>SK: Item Number</i>
Normal Attributes: <i>Reference Number - Item Number - TransactionID - Status - Agent ID</i> <i>Declarant CR - Declarant Name - Consignee CR - Consignee Name</i> <i>Exporter CR - Exporter Name - Commercial Description - HSCode</i> <i>HSCode_8 - CP3 - Regime - Customs Office Code</i> <i>Customs Office Name - LOC Code - Procedure - Package Amt</i> <i>Package Code - Sup Amt - Sup Unit - Terms of delivery</i> <i>Invoice Amount - Invoice Currency - Local Amount - Local Currency</i> <i>Local_to_Weight_Ratio - Gross Weight - Net Weight - Customs Duty Rate</i> <i>Customs Duty BHD - Excise Duty Rate - Excise Duty BHD - VAT Rate</i> <i>VAT BHD - Total Duty BHD - Fees - Country of Origin</i> <i>Country of Origin Code - Country of Export - Country of Export Code - Country of Destination</i> <i>Country of Destination Code - Registration Date - Registration Time - Registration Number</i> <i>Registration Serial - Receipt Date - Receipt Number - Receipt Serial</i> <i>Exit Date - Exit Time - Exit Number - Exit Serial</i> <i>Exit Office Code - Exit Officer ID - INDEX – Year</i>
SageMaker result attributes: <i>AnomalyScore</i> <i>SHAP_%_Impact Fields</i>
<i>SHAP_%_<Attribute Name> → SHAP impact value</i> <i>(Stored for many attributes listed above)</i>

Table 3. Data Dictionary of the Transaction Table

Table Name: <i>EmployeeActivities</i>	Attributes: <i>EmployeeName - Action</i> <i>ErrorExplanation - Flager</i> <i>item_number - reference_number</i> <i>TypeOfError</i>
Primary and Sort Keys: <i>PK: reference_number</i> <i>SK: item_number#EmployeeName#Action#Timestamp</i>	

Table 4. Data Dictionary of the EmployeeActivities Table

Chapter 5: System Design

5.1 User Interface Design

User Interface (UI) design focuses on the appearance and feel of a system with which users interact. It encompasses the layout, colors, fonts, buttons, icons, and overall visual design. The purpose of UI design is to develop an interface that is simple to use, visually appealing, and allows users to execute activities more effectively.

5.1.1 Login Screen



SenseAI
Login

Sign in to your account

✉ Email

🔒 Password

Sign in

Figure 2. Login Screen

Figure 2, the login screen displays the website's login credentials for auditors. A form must be completed after being taken to the login screen. The auditor must provide their email address and password; if they are not authorized, they will not be able to log in.

5.1.2 Home Page

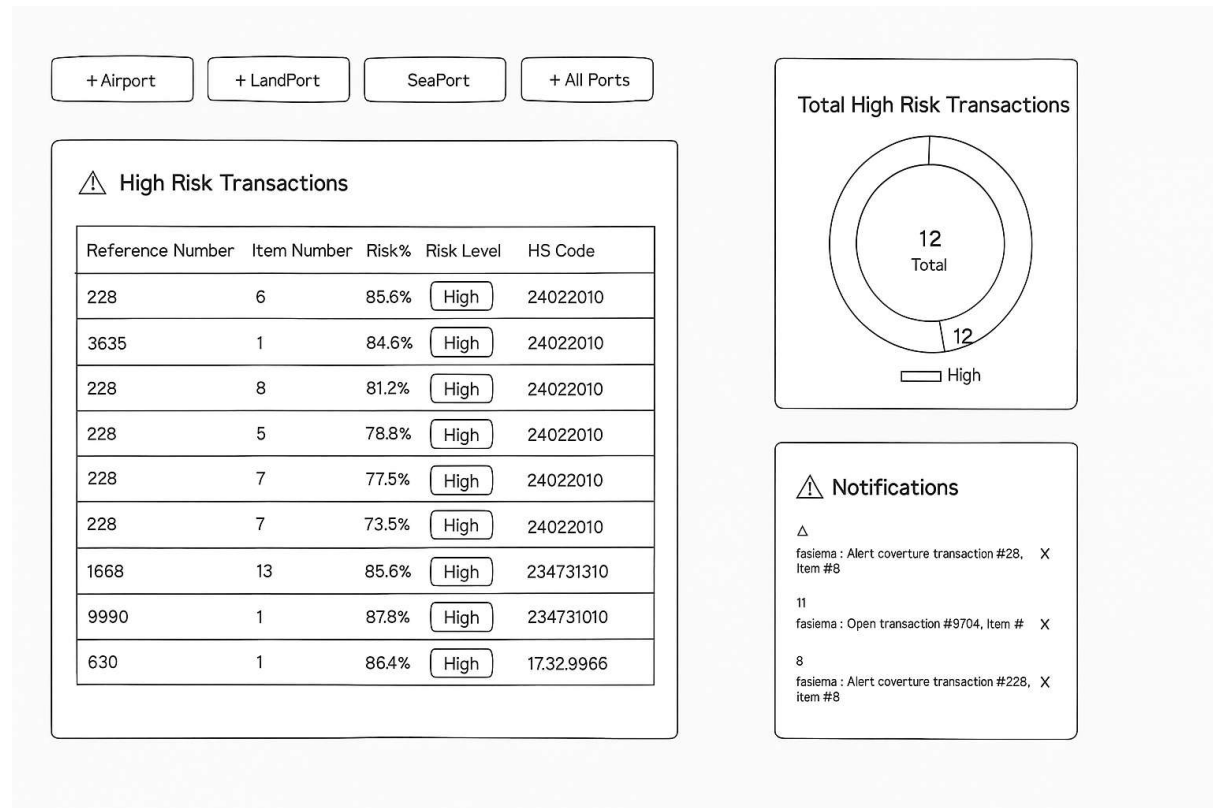


Figure 3. Home Page Screen

Figure 3, once auditors successfully log into the system, they are presented with a user-friendly dashboard highlighting high-risk transactions. The core table includes five key columns: Reference Number, Item Number, Risk Percentage, Risk Level (Critical and High only), and HS Code. Supporting visual insights, a pie chart displays the distribution of high-risk transactions. Real-time notifications alert users to important actions taken on the platform. Additionally, the dashboard offers port location filters, Airport, Landport, Seaport, and All Ports, enabling focused analysis based on entry points.

5.1.3 Upload Page Screen

The screenshot displays a web interface titled "Upload Files". Inside this container, there is a sub-section for selecting a date. It features a "Year:" label followed by a dropdown menu currently showing "2025". Below the year selector is a 3x4 grid of buttons representing the months of the year: Jan, Feb, Mar, Apr in the first row; May, Jun, Jul, Aug in the second row; and Sep, Oct, Nov, Dec in the third row. To the right of this calendar interface is a button labeled "Upload New File".

Figure 4. Upload Page Screen

Figure 4, this screen allows auditors to upload transaction files that need to be analyzed. It includes a calendar interface where users must select both the year and month before uploading, to prevent duplicate submissions, only one file is allowed per month. If a file already exists for the selected month, the user is prompted to choose between two options: Override, which deletes the existing file and replaces it with the new one, or replace, which updates only the modified portions of the original file. The interface also features an Upload File button, which enables the user to select and upload a file. All uploaded files are securely stored in an Amazon S3 bucket for further processing.

5.1.4 Transaction List Analysis result screen

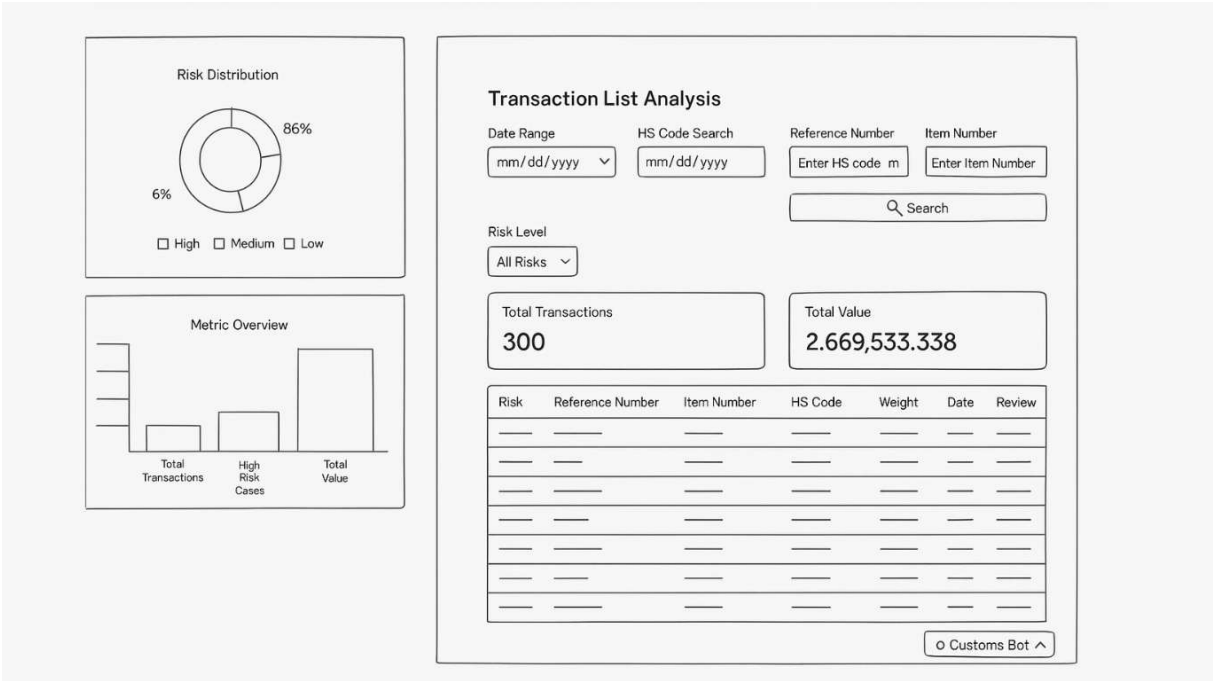


Figure 5. Transaction List Analysis Result Screen

Figure 5, this screen displays the transaction analysis results generated by the AI system. The results are presented in a table with seven columns: Risk Percentage, Reference Number, Item Number, HS Code, Weight, Date, and Review. Users can interact with built-in filter functionality to narrow down the results based on date range, HS code, reference number, item number, and risk level. At the top, the screen also highlights the total number of transactions and total transaction value in Bahraini Dinars (BD). Two visualizations support further analysis: a pie chart showing the percentage distribution of transactions across four risk levels, Critical, High, Medium, and Low, and a bar chart that provides a metric overview of the total number of transactions, high-risk cases, and overall value. These tools help auditors quickly interpret risk patterns and prioritize follow-up actions.

5.1.5 Transaction detail screen

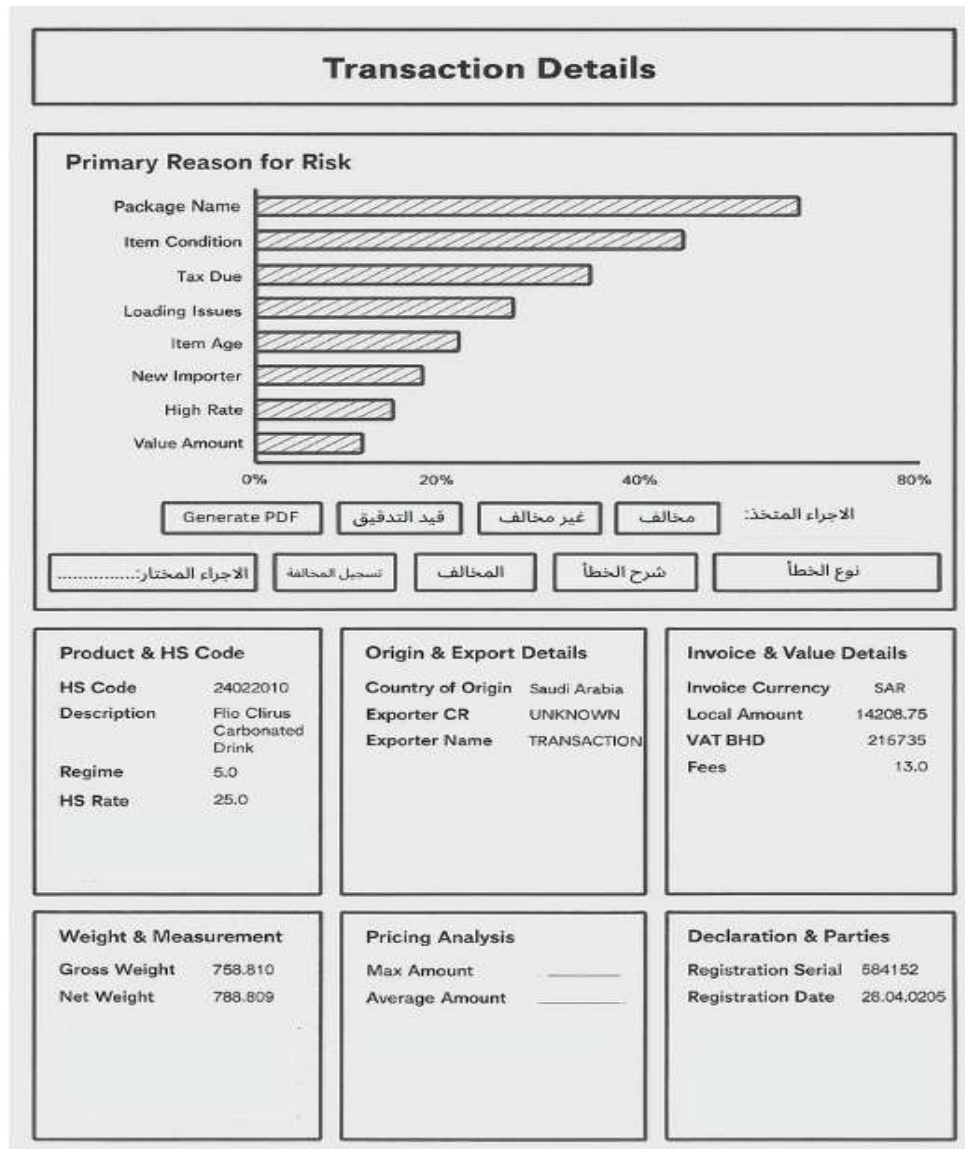


Figure 6. Transaction Detail Screen

Figure 6, this screen presents the detailed view of a single transaction, allowing auditors to review all relevant data extracted from the uploaded file. Users can take specific actions on the transaction, such as flagging it for violations, specifying the type of error, the reason for the violation, and identifying the responsible party. All actions performed are recorded and stored as activity logs for future reference. Additionally, the screen includes a visual chart that highlights the contributing factors behind the violation percentage, helping users understand the basis of the risk score. The interface also supports the generation of a PDF report that summarizes all transaction details and actions taken, making it convenient for documentation and audit purposes.

5.1.6 Employees Activities Page

	Date ▾	Action ▾	Employee ▾	 Search Transaction	Generate pdf
reference number	item number	Employee	Date	Action	Status
124	1	Fatima	11-5-2019	Open	Completed
880	2	Shereen	11-5-2019	Open	Flagged
228	7	Ahmed	11-5-2019	Open	Completed

Figure 7. Employees Activities Screen

Figure 7, the auditors will be provided a detailed dashboard that enables them to track the actions done by employees on the system as soon as they access the Employee Activities page. The page has a search box to find a certain reference number quickly, as well as filter controls by date, action type, and employee name. Essential fields, including reference number, item number, employee name, date, and action taken, are displayed in every record. Users can also export the filtered findings as a PDF report for use in audits or additional documentation.

5.1.7 Chat Bot Page

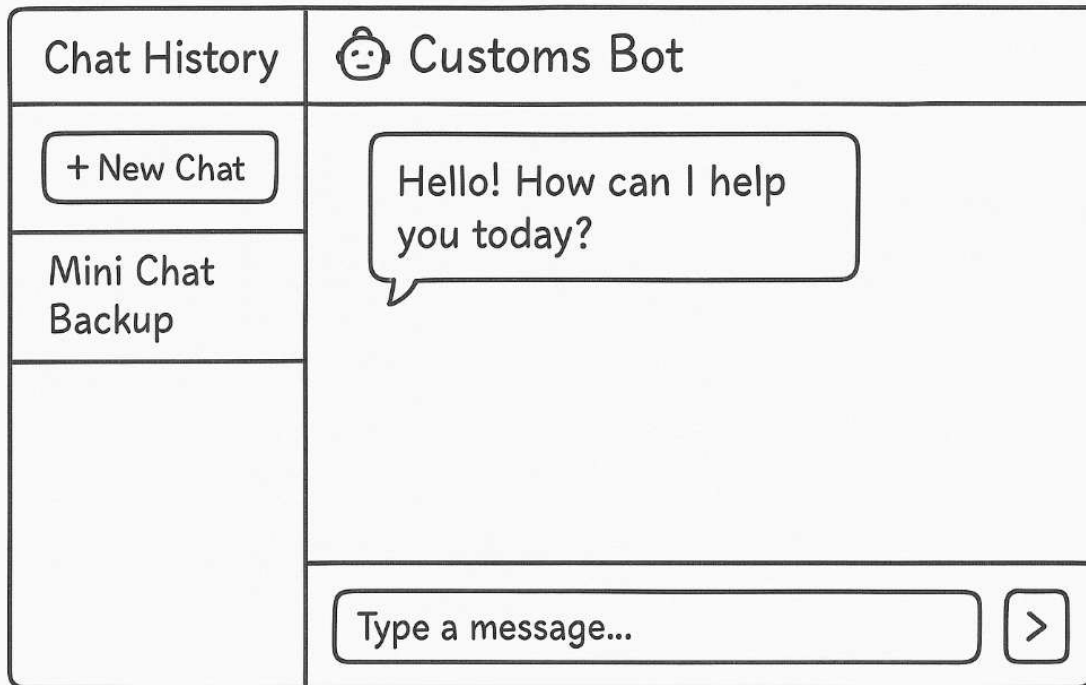


Figure 8. Chat Bot Screen

Figure 8, the auditors are provided with an easy-to-use interface that is intended to enable rapid and effective interactions as soon as they arrive at the Customs Chat Bot website. Users can initiate a new chat or see past ones, including saved backups, using the Chat History window that appears on the left sidebar. A real-time chat window with the Customs Bot is located in the main section. It begins with a welcome message that asks for user questions by default. Auditors can type their queries into a message input form at the bottom of the interface, then click a button that is clearly designated to send the message. This configuration facilitates smooth communication and makes it possible for auditors to quickly and effectively get information or help regarding customs procedures.

5.2 System Flow

Using a flow diagram, we will illustrate and describe the system usage process and procedure in this section.

5.2.1 System Flow Diagram

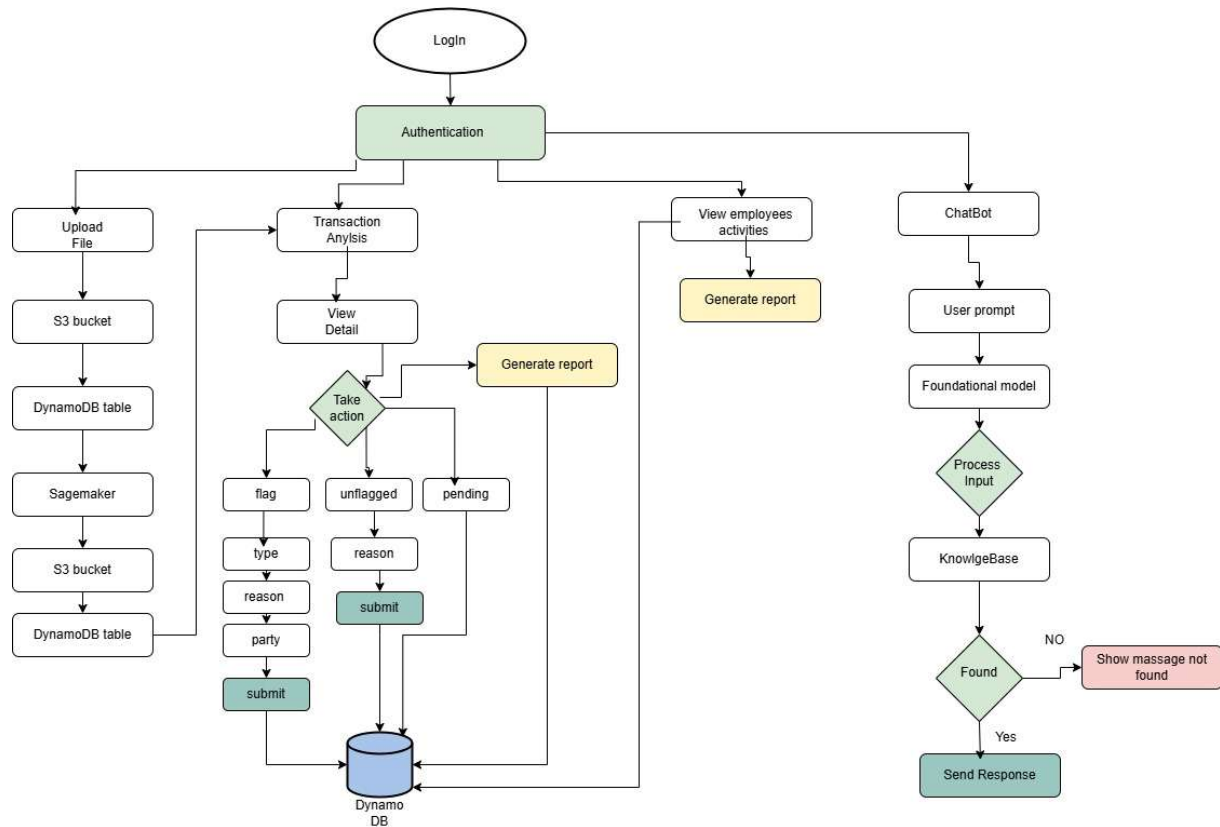


Figure 9. System Flow Diagram

5.2.2 Authentication

The authentication process, which is the first step in using the system, makes sure that only people with permission can access it. This entails using legitimate login credentials and using Cognito authentication to confirm the user's identity. The user can access pages and features of the system after authenticating.

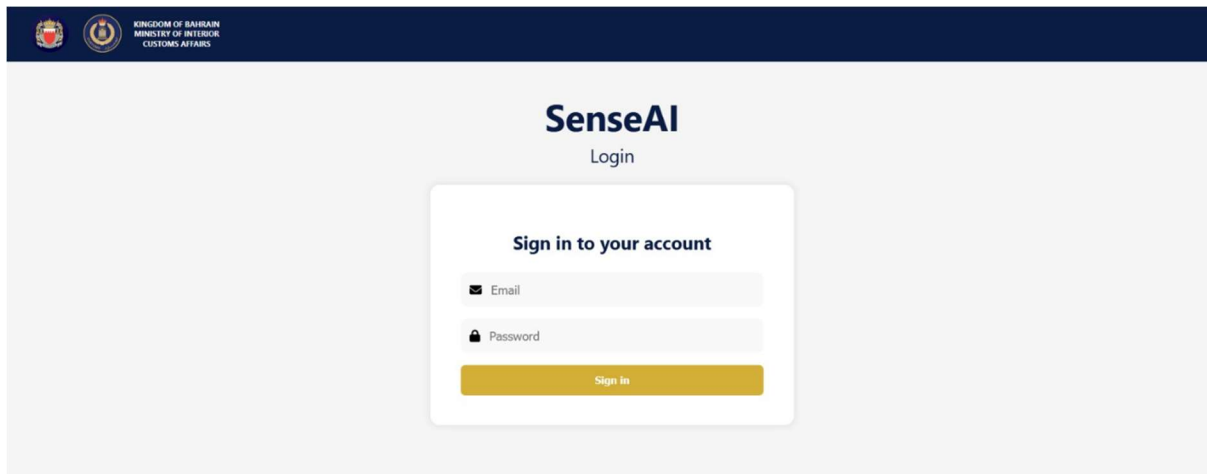


Figure 10. Authentication Screen

5.2.3 File Uploading

Users can access the file upload page following a successful authentication process. By choosing a certain month and year, users can upload transaction files here. The system offers choices to replace or override existing files and stops duplicate uploads for the same amount of time. After being uploaded, the data are safely kept and timestamp-tracked to guarantee that audit trails are preserved for later processing.

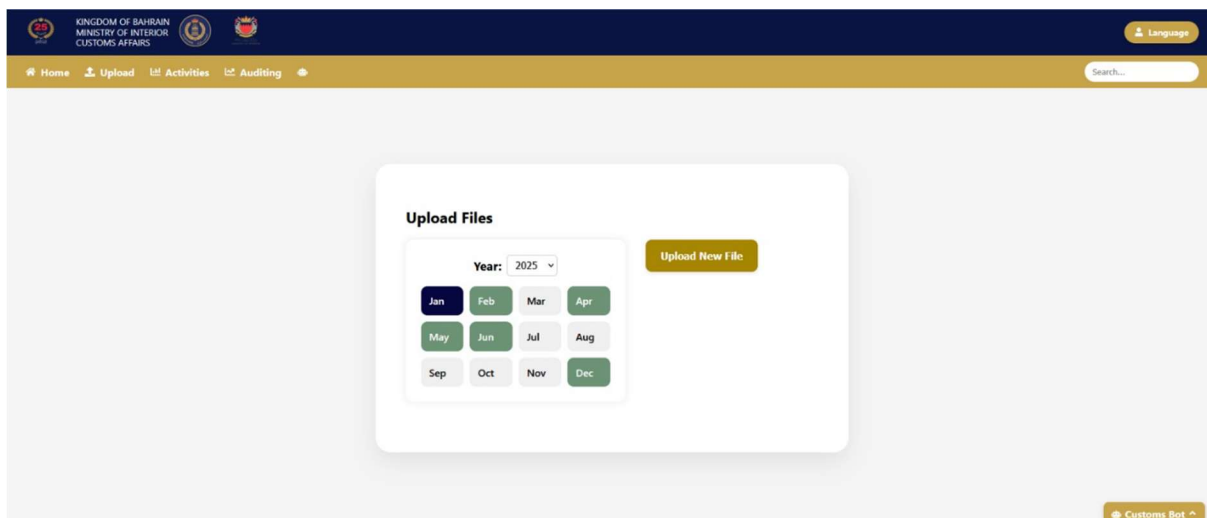


Figure 11. File Upload Screen

5.2.4 Auditing Page

The main dashboard for auditors is the auditing page. It presents a graphic representation of transaction data, complete with trends, risk categories, and summary metrics. Auditors can use interactive charts and tables to examine high-risk transactions and filter transactions by date, HS code, or risk category.

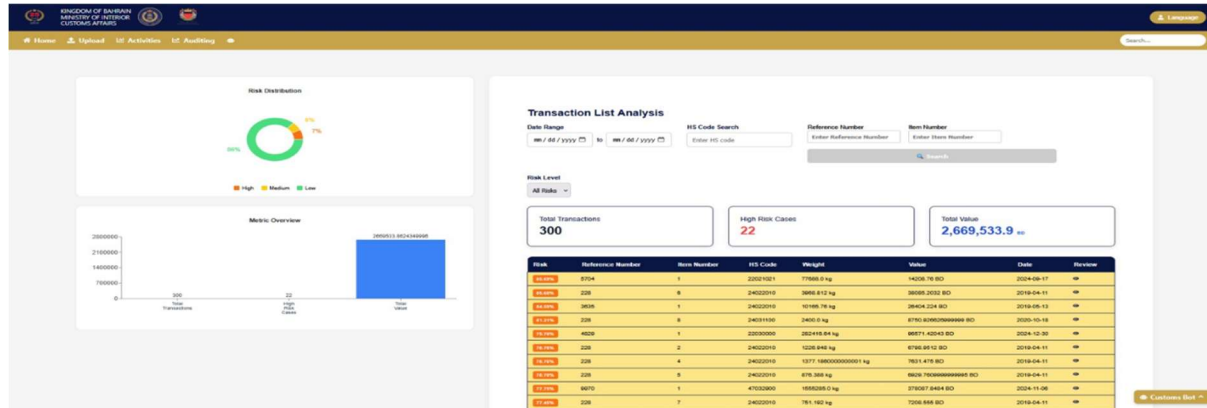


Figure 12. Auditing Screen

5.2.5 Transaction Detail View

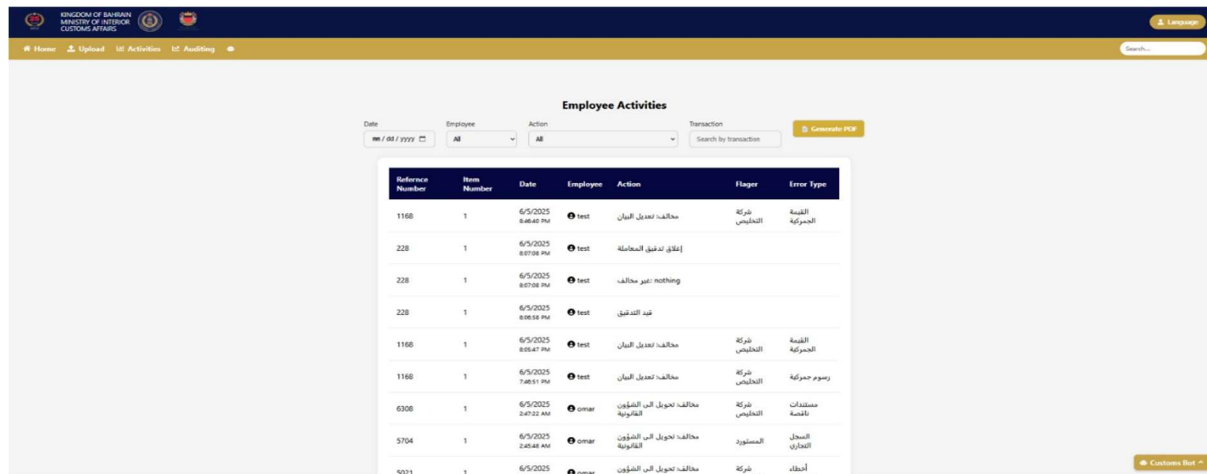
Auditors are directed to the transaction detail view when they click on a transaction review button in the audit table. Every field that was filled out in the original file is included in the detailed information about the chosen transaction that is displayed on this screen. A risk analysis graphic that highlights the contributing elements to any breaches that are identified is also included. Users have the option to flag, close, or comment on the transaction, among other actions. To ensure traceability and accountability, every action is recorded. Additionally, this page makes it possible to export filtered data into reports, which aids in documenting.



Figure 13. Transaction Detail View Screen

5.2.6 Employee Activities Page

The employee activities page is used to track individual employee actions and keep an eye on system usage. Logs can be filtered by reference number, date, action type, or employee name by administrators. Important metadata like timestamps and action labels are recorded in every entry. This tool facilitates performance reviews and internal audits while guaranteeing complete openness. Additionally, for documentation purposes, you can create reports of the complete list.



The screenshot displays the 'Employee Activities' interface. At the top, there's a navigation bar with 'Home', 'Upload', 'Activities', and 'Auditing'. Below this, a search bar and a 'Generate PDF' button are visible. The main content area features a table with the following data:

Reference Number	Item Number	Date	Employee	Action	Flag	Error Type
1166	1	6/5/2025 9:46:40 PM	test	معالجة تعديل البيان	حركة النظام	القيمة المبركة
228	1	6/5/2025 9:07:08 PM	test	إغلاق تدقيق المعاملة	حركة النظام	
228	1	6/5/2025 9:07:08 PM	test	غير معالجة: nothing	حركة النظام	
228	1	6/5/2025 9:08:58 PM	test	قيد التدقيق	حركة النظام	
1166	1	6/5/2025 9:08:47 PM	test	معالجة تعديل البيان	حركة النظام	القيمة المبركة
1166	1	6/5/2025 7:48:51 PM	test	معالجة تعديل البيان	حركة النظام	رسوم جمركية
6306	1	6/5/2025 2:47:22 AM	omar	معالجة تحويل إلى الشؤون القانونية	حركة النظام	مستندات ناقصة
3704	1	6/5/2025 2:45:48 AM	omar	معالجة تحويل إلى الشؤون القانونية	المستورد	السلع الواردة
5021	1	6/5/2025 12:47:47 PM	omar	معالجة تحويل إلى الشؤون القانونية	حركة النظام	البيان

Figure 14. Employee Activity Screen

5.2.7 Chatbot Interaction Page

Auditors can communicate with the Customs Bot, a virtual assistant designed to respond to questions about customs, via the chatbot page. A message pane, a real-time answer section, a new chat, and a history tracker are all features of the interface. Users can examine previous chats, start new ones, and ask questions on classifications, procedures, or transaction rules. This solution offers immediate, AI-driven support, improving user support and lessening the pressure on employees.

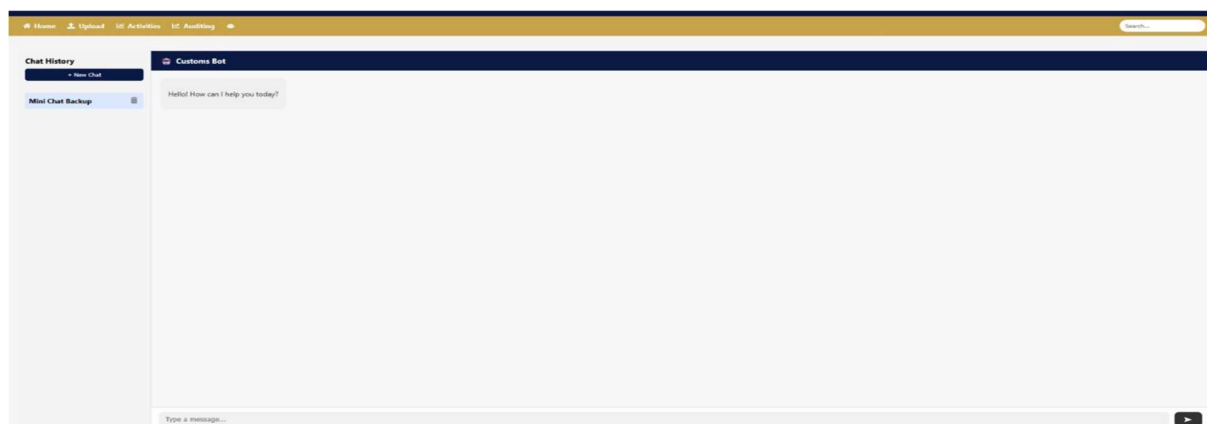


Figure 15. Chatbot Interaction Screen

5.3 Architecture Design

This section demonstrates how AWS services were used in our design to achieve the intended solution. Cloud architecture is the design of systems to be simultaneously scalable, reliable, and efficient; it provides services in the context of cloud computing. Architects would have access to a range of cloud services for networking, infrastructure, security, and database applications through AWS (Amazon Web Services). These flexible and affordable solutions subsequently made life easier by enabling businesses to create and implement systems that could now develop into what they need to be.

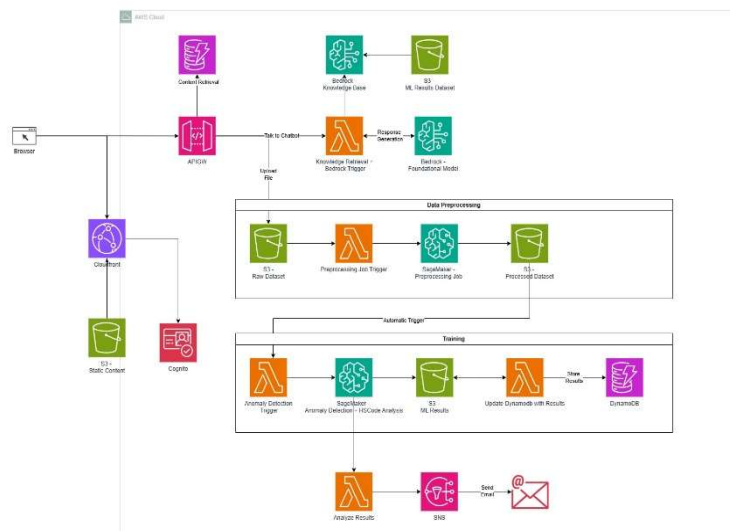


Figure 16. Architecture Design

Figure 16 illustrates the system architecture, encompassing the authentication process and access to the API. We will elaborate on each component of this architecture.

5.3.1 Portal Architecture

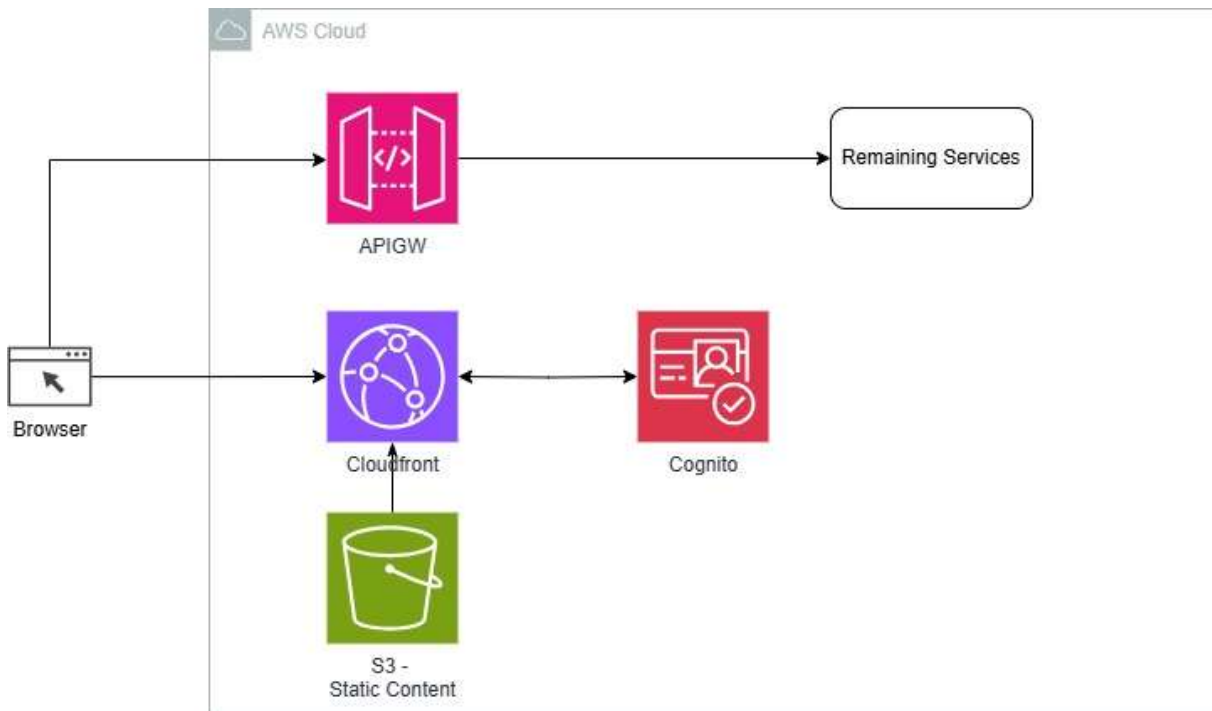


Figure 17. Portal Architecture

Figure 17 shows the architecture of the web portal where users interact with the website. The frontend is hosted in an S3 bucket and delivered through CloudFront, ensuring fast access and scalability. Authentication is handled by Cognito, which securely manages logins and issues access tokens. These tokens allow the frontend to communicate with backend services via Amazon API Gateway, where the API gateway verifies these tokens and routes requests to various services such as Lambda functions, databases, and analytics engines.

5.3.2 AI Chatbot Architecture

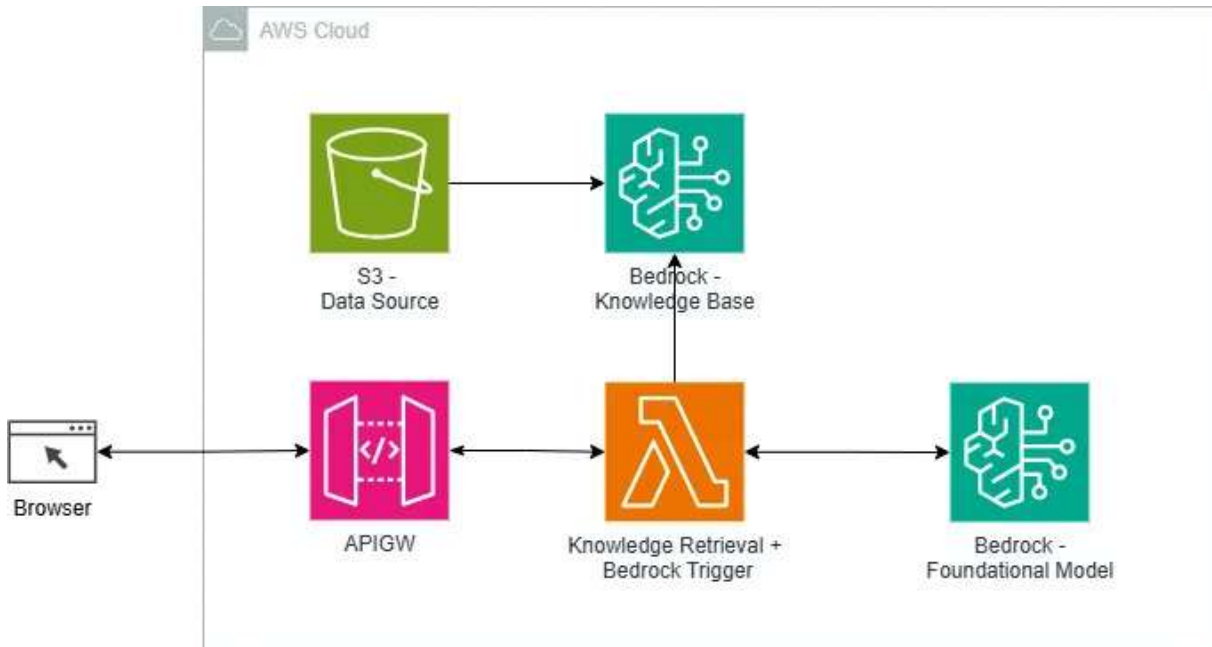


Figure 18. AI Chatbot Architecture

Figure 18 shows the architecture of the AI-powered chatbot process using Amazon Bedrock. The chatbot interface is accessed through the browser, which communicates with backend services and processes user input via the Amazon API gateway. Where the API gateway routes the user prompt to an AWS Lambda function, which is responsible for knowledge retrieval. The lambda function initiates a Retrieval-Augmented Generation (RAG) process by interacting with Amazon Bedrock. The query is first matched against a bedrock knowledge Base, which contains two types of content documents manually uploaded by the customs team, and machine learning results datasets stored in an Amazon S3 bucket. Relevant chunks are retrieved from the knowledge base and passed to a foundation model (Nova Pro v1.0) hosted on Bedrock. The model uses the retrieved context to answer the user prompt, which is returned to the browser interface.

5.3.3 Anomaly Detection Architecture

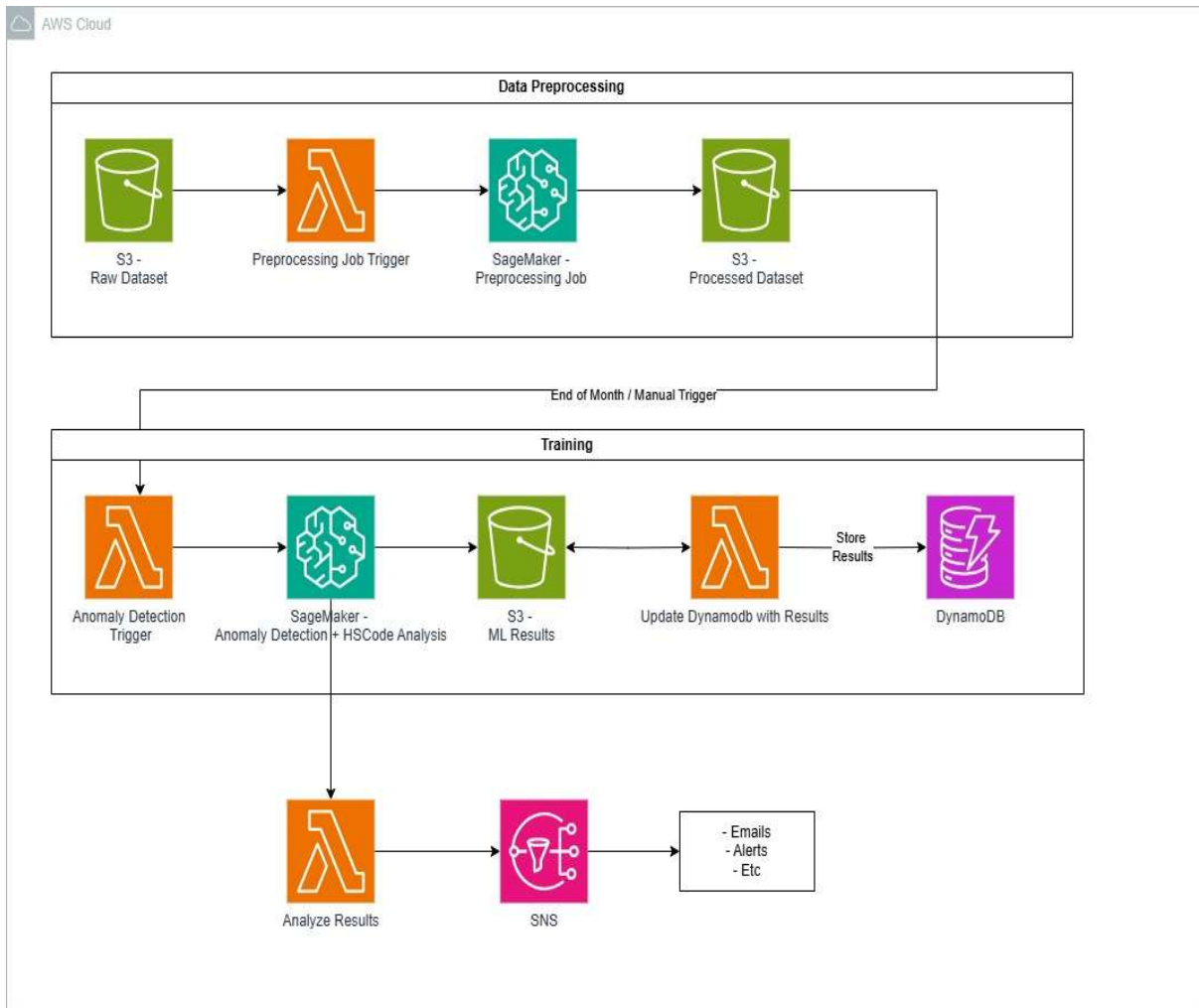


Figure 19. Anomaly Detection Architecture

Figure 19, shows the machine learning pipeline architecture used for customs anomaly detection, which is divided into two main stages: Data Preprocessing and Training & Inference.

5.3.3.1 Data Preprocessing part

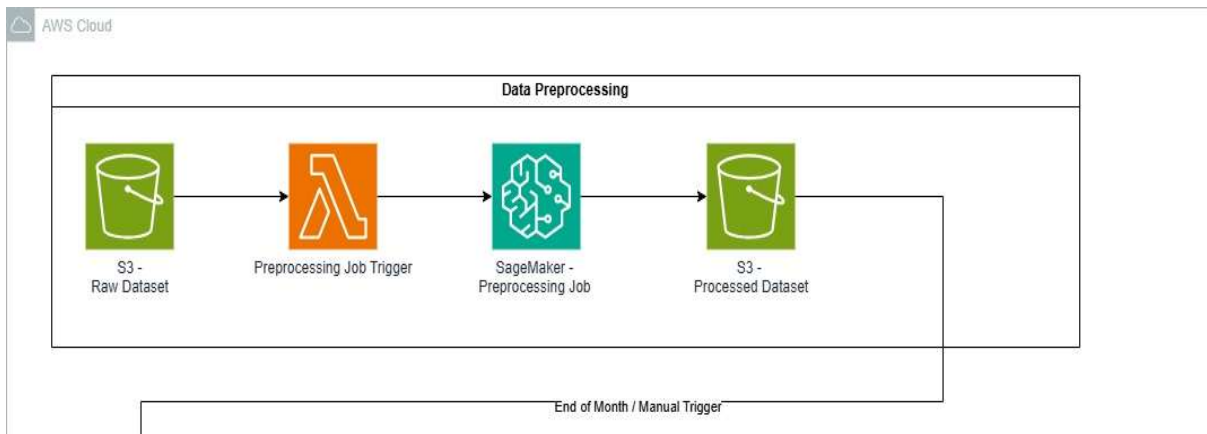


Figure 20. Data Preprocessing

Figure 20, this architecture shows data preprocessing. In this stage, a raw file is uploaded by the user and stored in an S3 bucket. A lambda function is invoked to initiate a SageMaker preprocessing job, which uses a defined script to clean and prepare the data. The output is stored in an S3 bucket (processed dataset) for further analysis. This output is later merged with previous data to build a complete training dataset.

5.3.3.2 Training and Inference part

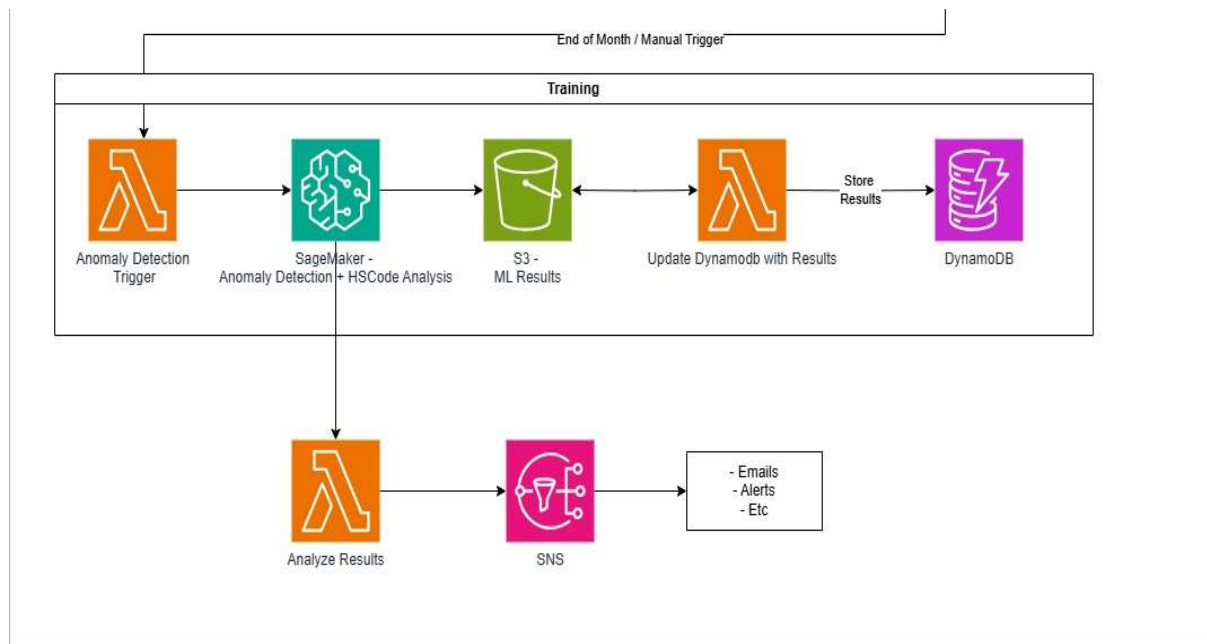


Figure 21. Training and Inference

Figure 21, this architecture shows the training part. In this stage, at the end of each month or when a new dataset is stored in the bucket, it will trigger a lambda function that runs a Sagemaker processing job that runs anomaly detection and HSCode level analysis. The results are saved in an S3 bucket (ML results). The results are then extracted by another Lambda function (Update DynamoDB with results), which then inserts structured output data into a DynamoDB table.

At the same time, another Lambda function (Analyze results) looks at the detection results and sends alerts over Amazon SNS (Simple Notification Service). These alerts can be emails or alarms.

5.3.4 Report Generation Architecture

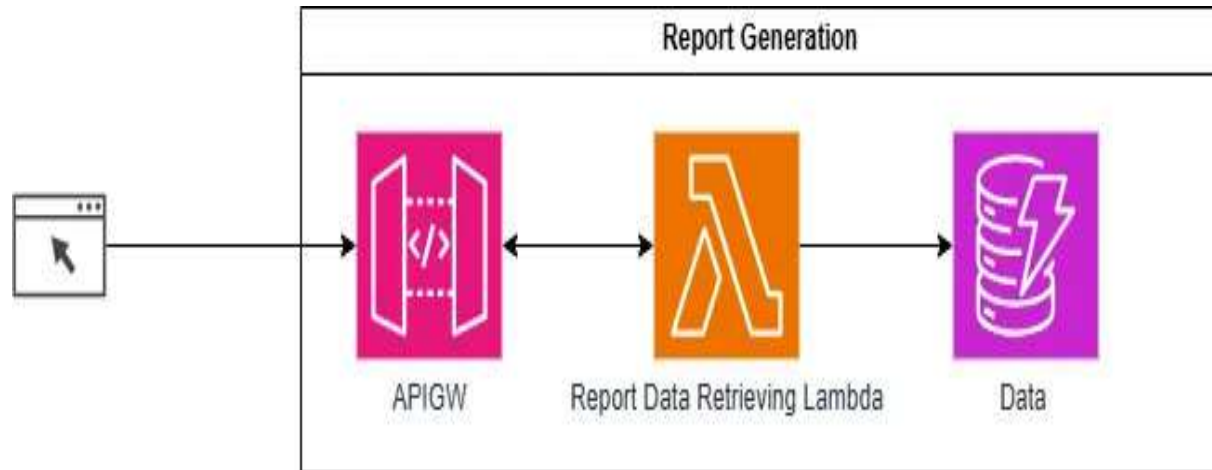


Figure 22. Report Generation Architecture

Figure 22 shows the report generation architecture. When a user initiates a report request through the browser interface, a request is sent via the API gateway to the backend. A lambda function processes requests and retrieves required data from DynamoDB. The retrieved data is then returned and displayed on the browser interface in a structured format.

5.3.5 Transaction Querying Architecture

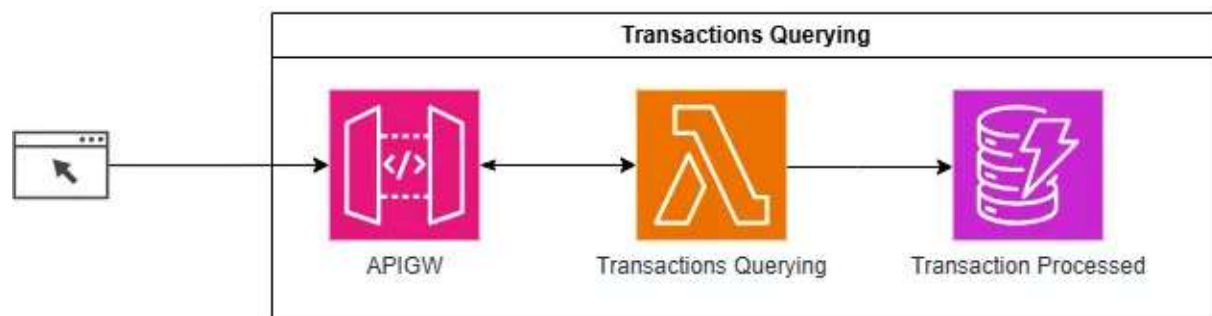


Figure 23. Transaction Querying Architecture

Figure 23 shows the transaction querying architecture. When a query is sent from a browser, the request is sent via an API gateway to the backend. Lambda function receives requests from the API gateway and processes these requests by fetching the relevant results from the DynamoDB table that contains structured transaction-level metadata. Then the results are returned via the API gateway to be displayed on the browser interface. This architecture enables dynamic and real-time lookup.

Chapter 6: System Implementation and Testing

This chapter presents the complete technical implementation and evaluation strategy of the SenseAI Customs Auditing System, a cloud-native, intelligent platform designed to support customs authorities in Bahrain with proactive anomaly detection, regulation guidance, and audit decision support. The system is built using Amazon Web Services (AWS), combining the power of machine learning with conversational AI and a secure, scalable infrastructure.

6.1 System Implementation

6.1.1 Software Tools Used

1. Amazon S3 (Simple Storage Service)

Purpose: Centralized storage for raw, processed, and result datasets.

What it does:

Amazon S3 is used to store all stages of our customs data lifecycle. This includes:

- Raw datasets uploaded by customs systems (monthly imports/exports).
- Preprocessed datasets are ready for model training or inference.
- Model outputs such as anomaly scores, SHAP values, and audit reports.

Why we used it:

S3 is secure, cost-effective, and scalable. It supports versioning, lifecycle rules, and seamless integration with other AWS services like SageMaker and Lambda. It's the backbone for data movement in our workflow, ensuring that data flows smoothly between preprocessing, training, and evaluation.

2. AWS Lambda

Purpose: Serverless automation and orchestration.

What it does:

Lambda functions are used to automate:

- Triggering SageMaker training or batch inference when new data is uploaded.
- Sending data between S3 and SageMaker or between SageMaker and DynamoDB.
- Calling the Bedrock chatbot endpoint for document-based answers.

Why we used it:

Lambda lets us run small, event-driven pieces of backend logic without managing any servers. It allowed us to build a highly modular, automated pipeline that responds instantly to data changes crucial for a live

customs auditing system.

3. Amazon SageMaker AI

Purpose: Training and deploying the anomaly detection model.

What it does:

SageMaker powers our core ML engine:

- Training: We trained an Isolation Forest model using scikit-learn on SageMaker managed infrastructure.
- Inference: Batch processing jobs score new customs transactions.
- Explainability: SHAP values are calculated in separate processing jobs for interpretability.

Why we used it:

SageMaker AI offers a fully managed, scalable environment to train and deploy machine learning models. With large datasets (~13 million records), it provided the computing power needed while reducing operational overhead. Using processing jobs also allowed us to parallelize SHAP computations efficiently.

4. Amazon DynamoDB

Purpose: Fast access to per-transaction insights.

What it does:

DynamoDB stores:

- Anomaly scores and SHAP values for each transaction.
- Audit metadata, such as flags, user comments, and review history.

The React dashboard queries DynamoDB in real time using Partition key: Reference Number (String) and Sort key: Item Number (String) to fetch relevant insights.

Why we used it:

DynamoDB is a low-latency, highly scalable NoSQL database perfect for storing and retrieving structured results quickly. It enables near-instant loading of audit data in the frontend without the delays of querying from a large CSV or relational store.

5. Amazon Bedrock (Nova Pro)

Purpose: Conversational AI assistant for auditors.

What it does:

Bedrock provides access to large language models (LLMs), and in our case:

- It powers a Retrieval-Augmented Generation (RAG) pipeline built with LangChain.

- The system retrieves relevant paragraphs from regulatory documents.
- Nova Pro (Claude) generates contextual answers to auditors questions.

Why we used it:

Bedrock offers secure, enterprise-grade access to advanced LLMs without managing the models ourselves. Nova Pro's performance and multilingual support made it ideal for answering questions about Arabic and English trade regulations in natural language.

6. Amazon CloudFront

Purpose: Fast and secure content delivery.

What it does:

CloudFront serves our ReactJS audit dashboard globally with:

- Caching for faster load times.
- SSL encryption for secure access.
- Reduced latency for users accessing audit pages from different regions.

Why we used it:

Customs auditors need responsive access, even when reviewing heavy data visualizations or SHAP charts. CloudFront ensures the dashboard remains fast and secure critical for field performance.

7. Amazon API Gateway

Purpose: Secure API endpoint management.

What it does:

API Gateway exposes:

- Backend APIs that return anomaly scores and SHAP values from DynamoDB.
- Bedrock chatbot queries (via Lambda).
- Audit action logs, flag submissions, and user authentication.

Why we used it:

It helps us control, monitor, and secure our APIs. API Gateway acts as a bridge between the frontend and backend, handling scaling, throttling, and access control automatically.

8. AWS Cognito

Purpose: User authentication and access control.

What it does:

Cognito manages:

- Login and identity verification for customs auditors.

- Role-based access control for dashboard features.

Why we used it:

We needed a robust and secure identity solution. Cognito integrates well with React and provides out-of-the-box support for password policies, multi-factor authentication (MFA), and user session handling.

9. AWS SNS (Simple Notification Service)

Purpose: Real time alerting and notifications.

What it does:

SNS sends email notifications to customs managers or supervisors when:

- High-risk transactions are flagged by the model.
- Manual flags are added by auditors for priority review.
- Model training or inference completes.

Why we used it:

Effective auditing relies on timely information. SNS helps keep stakeholders in the loop, ensuring that no high risk activity slips through unnoticed. Its integration with Lambda and email services made it easy to automate critical updates.

10. ReactJS (Admin Dashboard)

Purpose: Auditor-facing interface.

What it does:

The React dashboard allows customs officers to:

- Review anomaly scores and SHAP explanations
- Query the RAG assistant
- Flag, comment, or approve transactions
- Track audit logs and export reports

Why we used it:

React offers a fast, component-driven UI that works seamlessly with modern APIs and AWS services. It allowed us to build a clean, professional user experience that's both data rich and intuitive for non-technical users.

6.1.2 Implementation Overview

- Data Pipeline: Transaction data is uploaded, then stored in S3 and DynamoDB under a non-processed transactions table. A scheduled Lambda job sends it to SageMaker for pre-processing, followed by anomaly scoring.

- Anomaly Detection Engine: Isolation Forest in SageMaker scores each transaction based on deviation from historical patterns over the past 5 years of data that will be expanded with time.
- Audit Dashboard: ReactJS dashboard fetches scored data and visualizes risk levels (Low/Medium/High) with percentages.
- Conversational Assistant: Using Bedrock and RAG, the chatbot supports auditors by retrieving historical context, suggesting regulation clauses, or summarizing audit findings.

6.1.3 Algorithms Used

1. Isolation Forest (Unsupervised Anomaly Detection)

The Isolation Forest algorithm lies at the heart of SenseAI's ability to uncover potentially fraudulent customs declarations. Unlike traditional classification models that require labeled data, Isolation Forest is an unsupervised technique, making it ideal for real world customs data where labeled fraud cases are rare or incomplete.

In our implementation, the model was trained using the Scikit-learn library and deployed on Amazon SageMaker, which offered the scalability and compute performance needed to handle millions of historical trade records. The training utilized a balanced mix of numerical and categorical features, such as:

- HSCode (Harmonized System codes for goods)
- Local Amount (value declared in local currency)
- Gross/Net Weight
- Commercial Description
- Declarant and Exporter CRs
- LOC Code, and more.

These features help the model learn what constitutes "normal" trade behavior across different product types, regions, and entities. The model then isolates unusual transactions by randomly partitioning the data; transactions that are easily isolated are likely to be outliers.

Each transaction is assigned an anomaly score, which we scale to a 0 to 100% range for interpretability. A higher score suggests a greater deviation from learned norms. These scores are stored and later visualized in the dashboard to help auditors prioritize their focus.

This algorithm is fast, reliable, and well-suited for high-dimensional datasets, enabling us to detect fraud or compliance breaches without human bias or constant retraining.

2. SHAP (SHapley Additive Explanations)

While detecting anomalies is important, auditors need to understand why a transaction was flagged. This is where SHAP (SHapley Additive Explanations) comes in it breaks down the anomaly score for each transaction and shows which features contributed most to that score.

We incorporated SHAP as a batch inference step using a SageMaker Processing Job so it generates SHAP scores along with anomaly scores for each transaction processed. After the Isolation Forest model generates its predictions, a custom Python script calculates SHAP values for every feature in each transaction. These values represent the “impact” of a feature on the anomaly score, either increasing it (suspicious behavior) or lowering it (reassuring behavior).

To make these insights accessible, we store the top 10 most influential features per transaction in DynamoDB, indexed by a unique Partition key: Reference Number (String) and Sort key: Item Number (String). In the ReactJS dashboard, auditors can visualize these SHAP scores through an intuitive bar chart, where features are sorted by their absolute impact. This helps auditors answer critical questions like:

By offering per-transaction transparency, SHAP ensures the AI is not just a black box but a support tool that auditors can trust and justify.

3. Retrieval-Augmented Generation (RAG)

Auditors don’t just want to know what was flagged; they also need help understanding what to do next. For that, we integrated a Retrieval-Augmented Generation (RAG) system to support natural language audit inquiries.

Built using LangChain on top of Amazon Bedrock, our RAG assistant connects the dots between flagged transactions and relevant policies, manuals, and historical rulings. When an auditor asks a question such as:

“What documents are required to import electronic devices into Bahrain?”

“What are the requirements for customs exemption for personal luggage?”

“What documents are required for customs clearance?”

“What is the procedure in case of discrepancies between the customs declaration data and the actual goods?”

the system goes through the following steps:

- **Semantic Search:** It first searches the indexed policy and regulatory documents stored in S3 or other document repositories using vector similarity. Only the most relevant snippets are retrieved (e.g., duty rules, classification guidelines, exceptions).
- **LLM Reasoning with Claude (Nova Pro):** Using Amazon Bedrock’s Nova Pro model, the system crafts a human-readable answer, grounding the response in the retrieved documents and customizing it based on the auditor’s query.
- **Output to Chatbot:** The response is displayed directly in the audit dashboard’s chat window, complete with references and actionable suggestions.

This approach bridges the gap between raw anomaly detection and human decision-making. It transforms complex compliance material into practical insights, empowering auditors with the right information at the right time, without needing to dig through manuals or ask supervisors.

6.1.4 System Architecture and Infrastructure

The architecture consists of the following integrated layers:

- Data Layer: S3 buckets for datasets and results; DynamoDB for indexed SHAP outputs.
- Processing Layer: SageMaker Processing Jobs and Training Jobs handle transformations, training, and inference.
- Explainability Layer: SHAP values calculated during inference and stored in DynamoDB.
- Interaction Layer: React dashboard connects via API Gateway to Lambda endpoints.
- Assistance Layer: Bedrock chatbot uses a custom RAG chain to respond to auditor queries.
- Alert Layer: SNS sends automated email alerts for risky findings or job completions.

This architecture ensures:

- Scalability: Handles millions of rows efficiently.
- Security: IAM roles and data encryption are applied throughout.
- Auditability: Logs and DynamoDB entries offer traceability per transaction.

6.2 Evaluation and testing

This section discusses the testing methodologies applied to the SenseAI system to ensure robustness, accuracy, and secure performance.

6.2.1 Unit Testing

Unit tests were performed on individual components to verify their behavior in isolation.

Component	Functionality Tested	Result	Explanation
Data Preprocessing	Handling missing values, encoding categorical data	Pass	All critical rules were applied correctly
Anomaly Scoring Service	Returns a score between 0–100 based on Isolation Forest	Pass	Scores match expected ranges
Chatbot Query Processor	Returns valid contextual answers via Bedrock	Pass	Retrieved answers match relevant documents

Table 5. Unit Testing Table

6.2.2 Functional Testing

End-to-end scenarios were simulated to validate system functionality.

Test Case ID	Scenario	Expected Outcome	Result
TC001	Auditor uploads customs data	Data is stored, processed, and logged	Pass
TC002	Trigger anomaly detection	Scored output appears in S3 and UI	Pass
TC003	View transaction details	SHAP chart and metadata render in UI	Pass
TC004	Multiple users log in and audit simultaneously	Sessions handled without data loss	Pass
TC005	Query suspicious case via chatbot	Returns explanation and regulation reference	Pass

Table 6. Functional Testing Table

6.2.3 Integration Testing

We tested the interaction between major components to ensure end-to-end stability.

Scenario	Description	Expected Outcome	Result
SageMaker + React Web	Transaction scores appear on the dashboard	Live update in the chart and SHAP section	Pass
Bedrock + Document Store	Chatbot retrieves the correct reference documents	Accurate and contextual responses	Pass
Data Flow End-to-End	Full pipeline from data upload to audit insights	Seamless processing and output	Pass
SHAP + DynamoDB	SHAP values are stored and queried by the frontend	Accurate, per transaction display	Pass

Table 7. Integration Testing Table

6.2.4 Performance Testing

We evaluated the performance of core modules under normal and stress conditions. The system was evaluated with real production-scale data. The model inference testing was conducted on a subset of 99,000 rows, representing a realistic but manageable sample. The full dataset, containing over 13 million customs records, was used for training and historical analysis.

Function	Average Execution Time	Result	Scalability Notes
Model Inference + SHAP Scores per 99000 records (SageMaker AI)	8 minutes	Pass	Scales linearly with batch size
Training on the full dataset (13 million records)	23 minutes	Pass	Used ml.t5.4xlarge on SageMaker
RAG Query (Chatbot Response)	0.8 seconds	Pass	Slight latency increases under 5+ parallel queries
SNS Alert Trigger	~1 sec	Pass	Works reliably with multiple subscribers

Table 8. Performance Testing Table

This testing confirmed the system’s ability to handle real-world volume, deliver insightful results, and assist auditors without delays or bottlenecks.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

In SenseAI, we designed, developed, and deployed a cloud-based, AI-enabled auditing system to facilitate post-entry customs transaction analysis for the Kingdom of Bahrain Customs Authorities. The system is based on Amazon Web Services (AWS) and features a serverless, scalable architecture to automate the collection and processing, and visualization of transactions, allowing for risk-analysis. The system also includes a chatbot to allow officers to access regulatory/compliance knowledge and context using an AI-based knowledge base (KB).

At the heart of the system is a machine learning model deployed on Amazon SageMaker that provides each processed transaction with a risk percentage, which is considered one of the main and critical functionalities of SenseAI, and it indicates the likelihood of risk based on the transaction size and value, and many other metrics. This provided auditors with the ability to prioritize and provide focus on the transactions with the highest perceived risk. The machine learning model output is also enhanced with a form of explainable AI (SHAP values) to provide context to auditors to indicate the features that influenced the risk score, which makes their life easier in terms of what to focus on while auditing this huge number of transactions.

The key components that are part of the overall system include the following:

- A React frontend for officers to upload transaction records, view dashboards, view transaction profiles, and log officers' activities.
- AWS Lambda functions to enable the extraction and transformation of Excel-based files into structured records to be stored in DynamoDB
- A machine learning model hosted on SageMaker that provides a risk percentage to each transaction.
- DynamoDB tables are optimized for the performant storage and retrieval of transaction-based datasets at scale.
- Interactive dashboards to showcase risk ratings, trends, and allow for SHAP-based explanations of the features influencing the risk scores
- An AI-based chatbot that uses Amazon Bedrock to respond to officer inquiries utilizing a constantly evolving customs knowledge base.

The project was conducted with Bahrain Customs Authority and AWS Cloud Innovation Center (CIC) within a condensed two-month development time period and illustrated that cloud-native and AI technologies could improve post-entry auditing practices, support data-driven insight and improve efficiency.

7.2 Results and Findings

The project delivered a functioning system that:

- Processes big transaction datasets while maintaining scale and efficiency.
- Uses a machine learning model catalogued within Amazon SageMaker that assigns a risk percentage to each transaction for custom officers to assess and prioritize high-risk transactions.
- Provides dashboards for a customizable visualization and measurable insights into risk levels, emerging trends, and SHAP-based rationales for why each transaction got this score.
- Provides officers with support in the form of a multi-lingual AI chatbot that can answer questions based on official regulatory and explanatory documents.
- Provides audit logging and activity tracking for improved transparency and accountability of every action taken by an officer or auditor.
- Shows a secure and compliant deployment model consistent with Bahrain's IGA policy through a multi-region AWS structure.

Feedback from stakeholders has been positive regarding its overall efficacy. The system demonstrated a significant opportunity to reduce manual auditing time, enhance risk detection precision in audit practice, and provide greater transparency and explainability in customs audit processes. The iterative and collaborative development process allowed for the system to be appropriate to the actual operation.

7.3 Limitations

Several limitations were encountered during the project:

- **Time:**

The first week of April was the kickoff of our project development after the requirement collection phase, The team was limited to around two months to complete all the phases from designing to developing to testing the auditing system built with AI and complexity of AWS resources integration. Keeping this complexity in mind which included integrating many different AWS services, building a simple-to-use frontend, customized Chatbot with Retrieval augmented generation (RAG) and knowledge bases with Amazon Bedrock, developing AI-based knowledge retrieval with Amazon SageMaker, and processing large amounts of transaction data, this was a very short development time. The team had no choice but to build and focus on the essential features and manage delivery in an agile, iterative way. If they had been able to spend 6 - 8 months on the project, they could probably have built more advanced features, including better and more sophisticated machine learning models, to achieve deeper integration with customs systems, and to have tested the system more thoroughly in a real-world setting, to ensure robustness and future scalability.

- **Privacy Restrictions:**

Because this project involved working with extremely sensitive Kingdom of Bahrain customs data, protocols for privacy and confidentiality were followed throughout the development and testing of the systems. Expectations regarding data protection and compliance are explicit, with some types of information being withheld or redacted, specifically the details of past violations, names and identifiers of customs clearance companies, and sensitive operational activities. While the system is fully developed

and waiting for rollout in the Customs Authority, the lack of this data limited our ability to test some of the advanced features, such as entity-based risk profiling and violation trend analysis. These advanced features will be further tested, augmented, and validated once we are in the internal customs context where all the operational data can be used safely, and guidance and governance are properly in place.

- **Cross-Region overhead:**

Several key AWS services relevant to the SenseAI project, such as Amazon Bedrock, Amazon SageMaker, and Amazon QuickSight, are not supported in the Bahrain region (me-south-1). Due to its criticality to the system's functionality, critical services have been deployed to the Ireland (eu-west-1) region, which meets Bahrain's Information and Government Authority (IGA) terms of reference. Unfortunately, this meant that we had to implement a cross-region architecture, where the team was forced to place some related resources in different stacks, which required us to occasionally deploy separately in Bahrain and Ireland. While the SenseAI team was able to keep the system operational and compliant, this added some architectural complexity and cross-region data flow considerations. Some of the other non-essential services for the initially planned outputs were not implemented in this phase because it was not available for the Bahrain region. As an example, Amazon QuickSight, we believe, if it were supported locally, would have been a significant opportunity for additional value through data visualization and ease of implementation.

- **Data Dependencies:**

The system relies heavily on the data provided by the Customs Authority in order for us to develop, configure, and validate the feature set. During the life of the project, there have been multiple occasions where our journey was contingent upon receiving files and documents from the customs team. For example, there was the main transactional data set to populate the database and build our dashboards, documents, and regulations that had to be uploaded into the Bedrock Knowledge Base, as well as the formatting templates necessary for designing specific reports or visual displays. Oftentimes, the files had to be located, compiled, and cleaned, and only then could they be shared with us, which added natural delays to the project.

While understandable, given the sensitivity of the data that had to be processed and the various approval processes involved, it did affect our pace in coming through this phase at certain points. Working without this dependency could have improved the time efficiency in SenseAI's development cycle.

- **High Cost:**

Some AWS services and models were expensive and impacted some key architectural choices during the project. While our intent was to build a fully-featured and highly automated system, we had to forgo some specific capabilities due to cost, including advanced capabilities. For example, while AWS Translate would have provided easy multilingual capability for the user interface and chatbot responses and would reduce the manual time to translate our website, it was too expensive to use. Because of that, we needed to rely on manual and partial translation methods instead. Similarly, while Claude Sonnet is a high-performance and high-quality foundation model in today's AI world, its expensive price and necessity for approval to access it make it impractical for this project's phase. Instead, we opted for a variety of other

models with a fair price-to-performance ratio, considering and evaluating on cost throughout the design of the system.

- **Incompatibility:**

Some of the file formats and Amazon Web Services (AWS) presented a kind of challenge throughout the project. A key example of this was the customs team's use of legacy Excel documents. The parsing of the converted Excel documents occasionally throws errors or requires specific consideration upon uploading and parsing. Outside of Excel documents, the inconsistencies in behavior across the web browser in terms of how it uploaded files generated small annoyances, but these were manageable.

The bigger challenge with AWS was Glue. Glue was initially intended to be a way to evaluate the ability to do large-scale data processing. However, Glue does not natively ingest Excel files. The raw data from customs currently comes in an Excel format, and if we had followed the Glue path, we would have to process the Excel files in some way and require an additional set of steps to use Glue to convert the files to comma-separated values (CSV) format. The conversion step is intended to complicate an already complex process and increase the chance of data integrity concerns. For these reasons, we did not pursue a deployment with Glue for this phase. Other file format compatibility considerations also influenced the tool and architectural decisions made, as it reinforced the need to align the technology choices and decisions with the very real data formats and needs of the business/stakeholder.

7.4 Future Work

Despite the solid planning and development of SenseAI, Numerous improvements can be made in future development cycles to improve the functionality, scalability, and alignment to the operation of the Kingdom of Bahrain Customs Authority.

The AWS team will continue to work with the Customs Authority to improve the system as it enters full production. The following ideas are identified as key priorities for future work:

- **Supervised Machine Learning:**

At this stage, the transaction data being uploaded to the system does not include labels indicating whether transactions are compliant with or in violation of customs regulations. In the next phases of using the system, as customs officers and auditors will visit the system frequently to perform their duties (i.e., flagging transactions, housing violations), the system will accumulate records that will be centrally maintained and centralized within the system with the processes. This will allow us to build supervised machine learning models based on real-world auditing decisions or actions. The more time the system is used, the data will build a useful data set of labelled transactions to develop machine learning models that can better identify potentially suspicious or higher risk transactions. Once trained and validated, the Customs Authority can then determine whether to build this into the current system to improve the ability to predict transactions, or not, if it is not useful for their use case.

- **Role-Based Access Controls:**

The intended changes are to create a role-based access control to better align the system to the operational structure and security requirements of the Customs Authority. The system will support multiple user roles, each with different levels of access and functionality based on their responsibilities. For instance, top

management users might be supplied with dashboards to provide strategic insight, while operational users may be restricted to viewing or auditing transactions within their scope of responsibility. The administrative users will have a separate interface to create user accounts and modify privileges. This overall approach will help ensure sensitive information and system functions are securely managed in line with the user's role.

- **Improved Designs & Graphs with Customs Dynamics:**

While the development of SenseAI was focused on delivering the critical functional aspect of the system, there are opportunities to enhance the user interface and user experience. Future development work will focus on making the system as closely aligned with operational workflows and the Customs Authority brand identity as reasonably possible. Enhancements may include the use of official logos, dynamic visuals, or a more logical layout of tasks pertinent to customs officers. Also, the alternatives to Amazon QuickSight will be explored and followed up to enable dynamic or real-time visual representations of trends and insights captured through system operation and AI analysis as potential application areas for the officers and stakeholders to gain insight into the performance, risk, and activities occurring.

- **Preview Records of Uploaded Files:**

As one of the later enhancements, SenseAI will come with a built-in file preview with edits within the upload routine system. As it stands, users need to access external systems like Microsoft Excel to review uploaded files and verify their contents. Future versions will provide functionality to preview uploaded data directly from the website itself through the upload page before ingesting it into the system, which will allow users to confirm that the data is accurate and complete, make changes as necessary, and complete the process without re-uploading files. In general, these improvements will expedite data validation within the system, improve the user experience, and minimize the chance of errors that occur while entering data.

- **Advanced Bedrock Knowledge Base:**

The Bedrock Knowledge Base will remain a work-in-progress feature of the system as it is being developed and improved. The upload page will facilitate the uploading of new regulatory documents, guidelines for processing, FAQs, and other components relevant at the time for the Customs team. These additions will ensure the knowledge base remains relevant and thorough as customs regulations and practices continue to change. Even sensitive documents and detailed processed documents that were not included during the initial development period can also now be uploaded. As a result, the AI-powered chatbot will now be able to provide advanced, contextually relevant, and comprehensive answers. For example, the AI-powered chatbot will be able to describe why certain transactions were flagged as extremely high-risk based on the evidence and supporting pieces as part of the recommendations. New knowledge base additions will also promote transparency for the customs team and improve contextual risk assessments, which can facilitate more informed decision-making.

- **Advanced File Validation Rules:**

Another point to consider for future development is strengthening existing file validation to ensure the basic file is accurate before importing it into the system for processing reasons. The system will include data cleaning and advanced validation rules to identify missing columns, inappropriate formats, duplicates, and other common data quality issues. Since the system will be able to support uploads to both the transaction dataset and the Knowledge Base, which will later highlight the need for strong security controls to ensure unauthorized, malware and bash files are not uploaded for the reason that these types of files are evolving, and it is very easy to cause serious damage to the system. While the system is only used internally and all user actions are logged, which results in the fact that it is not exposed to the public internet, additional improvements to file validation and security would help maintain the integrity of the system and ensure a healthy data quality. This would ensure the system has the highest degree of security and improved resiliency for much longer sustainability and long-term use.

7.5 Implications

This project has shown the potential and advantages of implementing new technologies, such as cloud-based and AI, to effectively transform post-entry customs auditing. The system offers the Kingdom of Bahrain Customs Authority important, scalable, intelligent, and explainable auditing capabilities that can tremendously improve operational efficiency, accuracy, and transparency.

At the core of the system is a machine learning model hosted on Amazon SageMaker that applies risk percentages to individual transactions to allow officers to prioritize their time on the highest risk cases. The addition of SHAP-based explainability to improve trust in the risk scores represents a profound step forward in the practical use of risk-based auditing at scale, as officers have a better understanding of the factors behind each risk score.

The system automates workflows, and provides AI-powered insights, as well as a multi-lingual Bedrock Knowledge Base chatbot to support officers by pulling the manual effort out of doing audits, and provides officers with a better understanding of the data to ultimately enable better data-driven decisions. Additionally, the architecture allows for the full traceability of audit processes through activity logs and the ability to comply with Bahrain's data governance because of the AWS multi-region architecture.

The project advances the Customs Authority's capacity to drive further digitalization of its operations. Additional features such as supervised machine learning, role-based access controls, advanced visual analytics, and a growing Knowledge Base will all add value and subsequent impact to the system. The system's modular architectural design will allow for the future demonstration and delivery of new capabilities and functionalities.

On a wider scale, this project highlights the Bahrain Customs Authority as a recognized regional leader in AI-powered customs auditing. This project serves as an example of how intelligent, explainable, and transparent AI solutions can be responsibly applied to the public sector setting to improve operational effectiveness while complying with governance frameworks and maintaining public trust.